

Computer Vision Software Development Procedure (For Use at RoboSub Competition)



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Introduction:

This document was created by the 2025-2026 *RoboSub Computer Vision* Senior Design Team, consisting of Luke Connolly, Keenan Murnion, Kye Buerkle, and Joshua Sapienza. The document was created for RoboCats usage, so that every member of RoboCats could follow this procedure, to create a working computer vision algorithm for the submarine, at the RoboSub competition.

In the creation of this procedure, the Senior Design (Capstone) Team split up the processes of the development of the computer vision software into four subsystems: Data Collection, Data Preparation, Model Creation, and Model Upload. Each member of the Capstone team was responsible for one subsystem throughout the Capstone course, creating reproducible processes for each part, culminating in this document, which hopefully serves as an instruction guide for RoboCats members.

The initial problem that the Capstone team was given was that the algorithm worked fine, even at competition, but it was difficult for members of RoboCats who were not familiar with the software to create (and thus tweak) the algorithm at competition. The knowledge on how to do it was not well documented anywhere. So, the Capstone team decided that the output of this project would be this procedure, serving as an instruction set for every member of RoboCats, so that all members can follow the steps in the creation of the computer vision software.

Another problem addressed by the Capstone team was the algorithm's performance in different lighting conditions at the 2025 Competition. The Capstone team addressed this problem by adding an image augmentation step in the algorithm development. Essentially, augmentations such as increasing and decreasing exposure values, contrast, and more are now applied to the training images (the images used to train the machine learning model). This allows the model to have a more diverse training set, improving detection accuracy under different lighting conditions at competition.

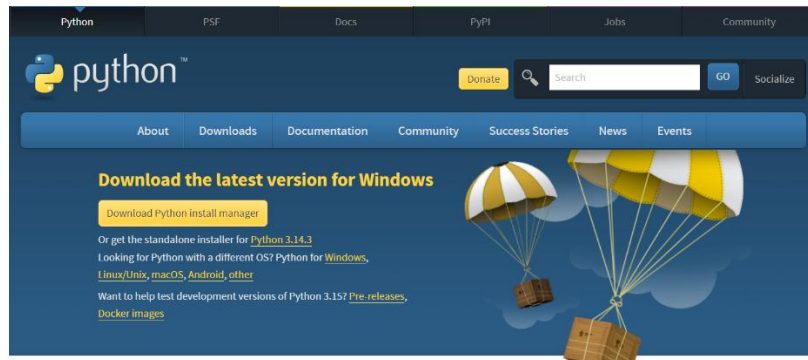
The procedure is split into two parts: Setup, and Competition Workflow. The Setup section immediately follows this introduction, detailing steps that need to be taken before competition, such as creating accounts, gathering materials, and preparing storage on your personal computer. It is recommended that the Setup section be completed well before competition, as some steps require account approvals that may take several days. The Competition Workflow section is a step-by-step guide meant for use at competition, and should be followed sequentially from Data Collection through Model Upload, as each subsystem builds on the output of the previous one.

Data Collection – System Setup

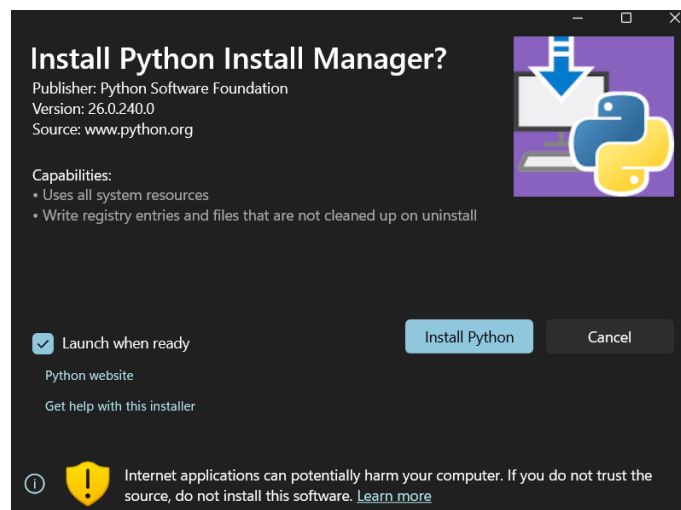
System Setup:

This setup is essential for the dataset curation subsystem to function in a timely manner and so that the dataset curation scripts will run on a laptop without previous experience using command terminals and python. This procedure assumes that the MicroSD card has already been formatted, the required RoboSub folder structure has already been created, and the data ingestion script has already been placed in Scripts_RBSTrain. If this has not been done, refer to the MicroSD card preparation procedure and the data augmentation setup procedure.

1. This section involves using Python, so the user must have it installed. To check if you do, open Windows PowerShell and type `python --version`. If a version number appears starting with 3, skip to step 12. If you get an error, continue to step 2.
2. Go to <https://www.python.org/downloads/>



3. Click on **Download Python Install Manager**
4. Open file explorer, and go to the downloads folder
5. Click on the file **python-manager-26.0**. This will open the install manager.
6. Click on Install Python



7. This will open the Python installation manager configuration helper.
8. You will then get prompted with two configuration questions. Press y to say yes to both of them.
9. You will then get another prompt question, for opening the help menu. Press n for no, and the install manager terminal will close.
10. Open Windows PowerShell on your device.
11. Type the command **python --version**. You should see what python version was installed on your device.
12. Now you need to install the required packages needed for the scripts to work.
13. Install OpenCV by typing the following command into Windows PowerShell: **py -m pip install opencv-python**
14. Install Pillow using **py -m pip install Pillow**
15. Install NumPy using **py -m pip install numpy**

Data Preparation – System Setup

System Setup:

The entire labeling process will take place on the Roboflow website, which can be accessed from any personal computer. Because of this, the materials needed to complete this process are minimal. However, for purposes of absolute clarity, using a laptop (instead of a smart phone or iPad) is mandated for this process. This is because the design and process flow were created while using a Dell laptop, and any subsequent errors or quirks associated with this type of interface were documented and thought of. Any irregularities associated with using an iPad or smartphone were not considered in the writing of this process. It should go without saying, but access to reliable internet is also a requirement. If hotel internet (at competition) is not suitable, a personal HotSpot is required.

To use the Roboflow website, you must have an account. The following is a step-by-step walkthrough of how to create an account on Roboflow.

1. Open Roboflow website on computer. Website domain is <https://roboflow.com/>
2. Your screen should look like Figure 1 (current version of Roboflow home screen). This image is to ensure you are at correct starting point.

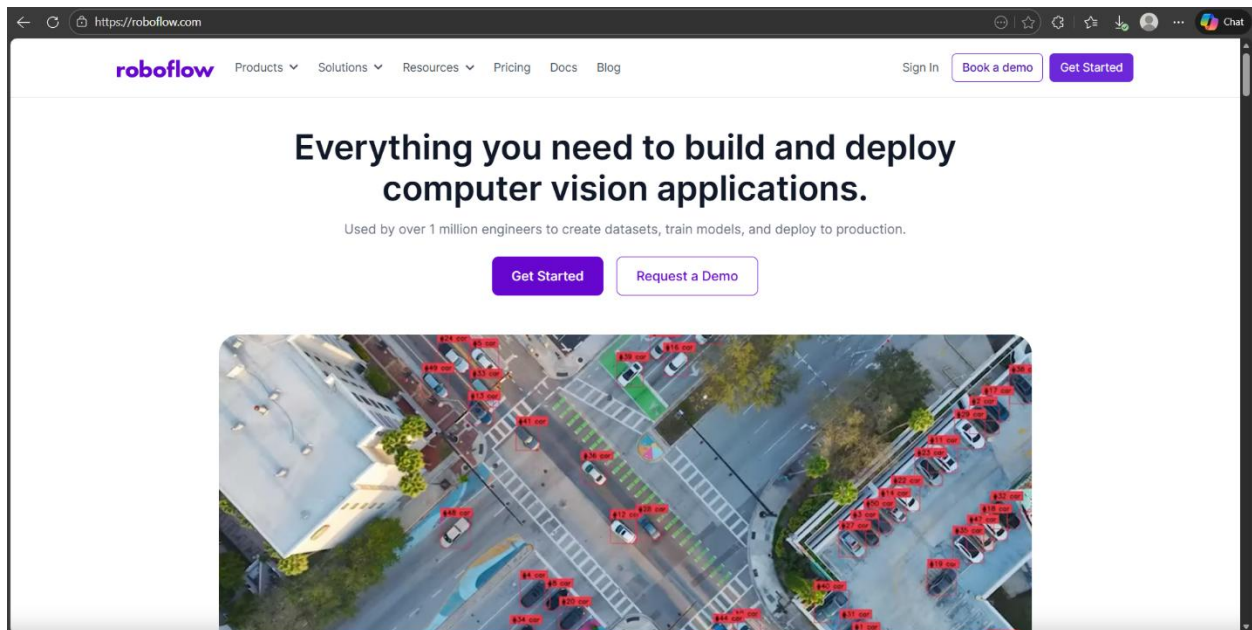


Figure 1. Roboflow Home Screen

3. Select the “Get Started” button, located in the top right corner.

4. This will take you to another page, asking you to “sign in or sign up” through either your google account, GitHub account, or personal email. Select the personal email option.
5. Enter your preferred email address. As you will later find out, Roboflow costs “credits” to complete some actions on the site. A normal free account currently gets allotted 30 credits/month, but an email associated with an accredited university gets 50 credits/month. Assuming a Montana State student is reading this, use your student.montana.edu email.
6. A confirmation email will be sent to your entered email address. Open the email and confirm access. After clicking the link in your email, you will be sent back to the Roboflow website to confirm account details.
7. Enter your preferred name when prompted. Accept the Terms and Conditions. You will be redirected to this page.

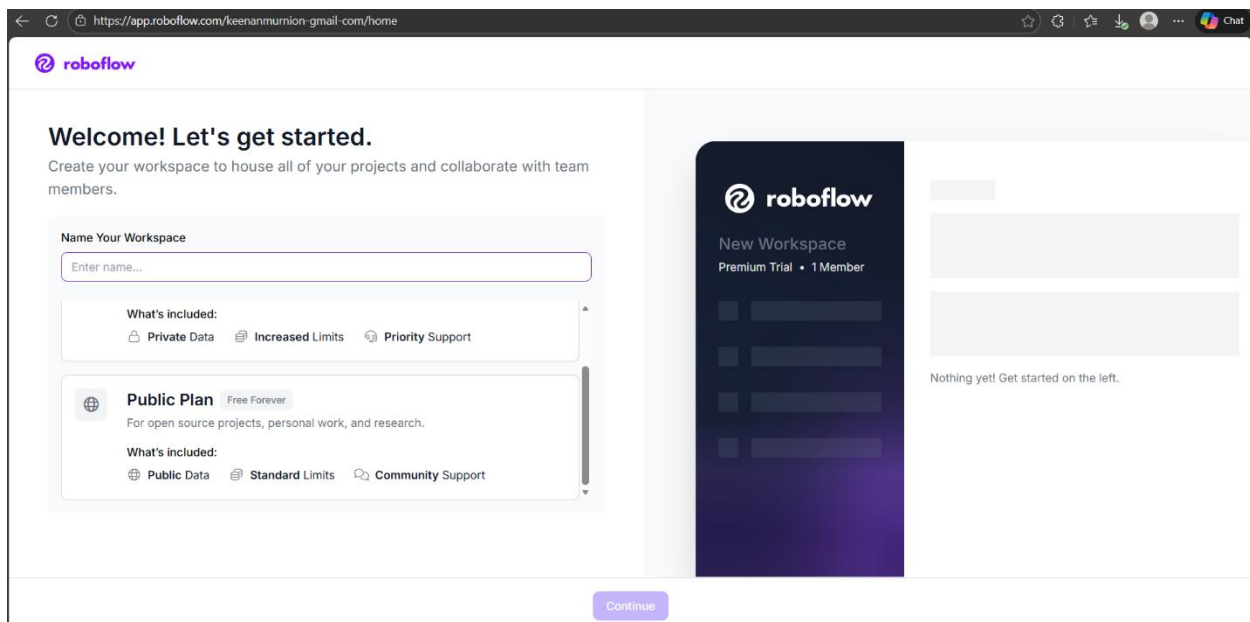


Figure 2. Roboflow Workspace Creation

8. Name your workspace appropriately. Your name could be “RoboCats 2026”, or something along those lines.
9. Two plans will be available: the Public Plan (seen in Figure 2) and the Private Plan. Select the Public Plan. (Public Plan is more than capable for our needs!). Select “Continue”.

10. The next page will give you the option to invite teammates to your workspace. You can use this page to enter your RoboCats teammates' email addresses, to invite them to the workspace.
11. Select "Continue" to fully launch your first workspace. Your screen should then look like this:

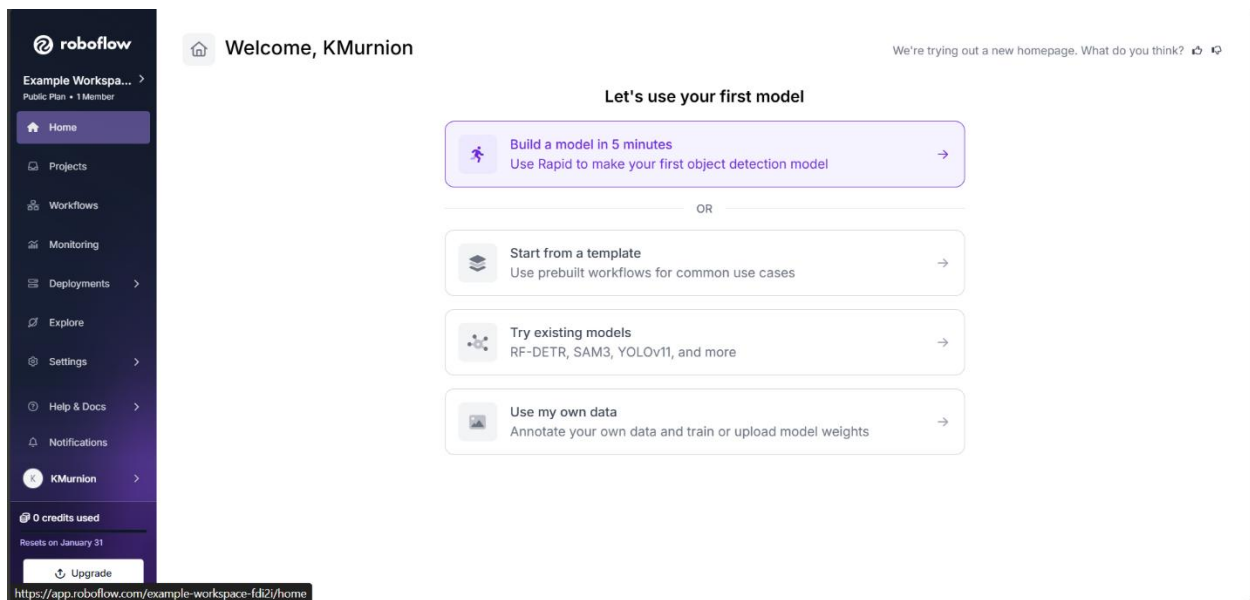


Figure 3. Example of RoboFlow Workspace

12. That is the final step in creating an account, as well as a workspace! Now, let's create a project before we go to competition.

13. Start a new project on the RoboFlow page. Figure 4 shows what your screen should look like.

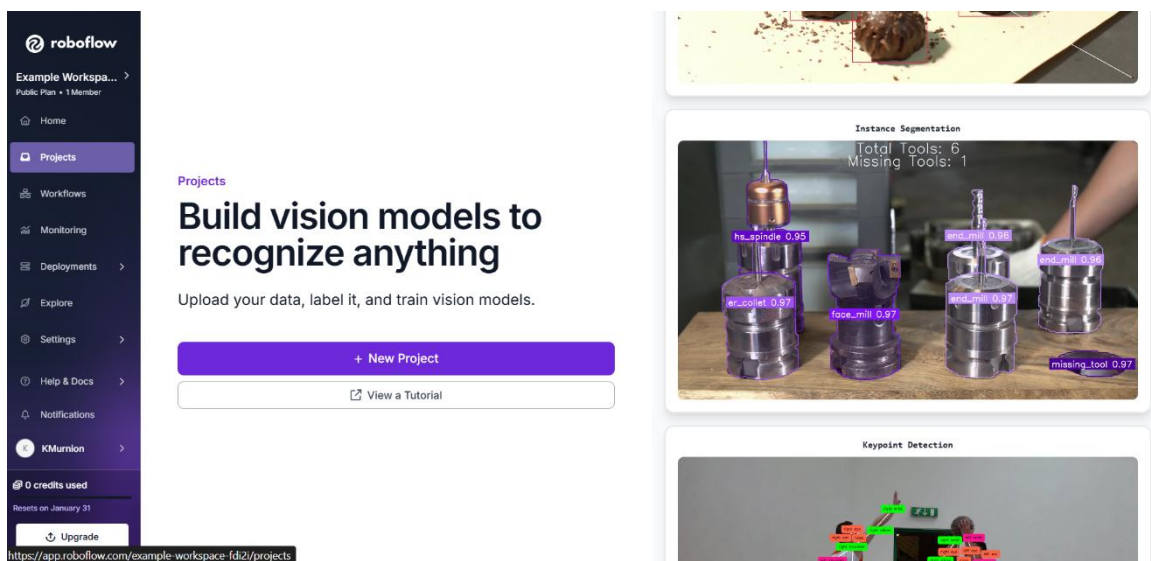
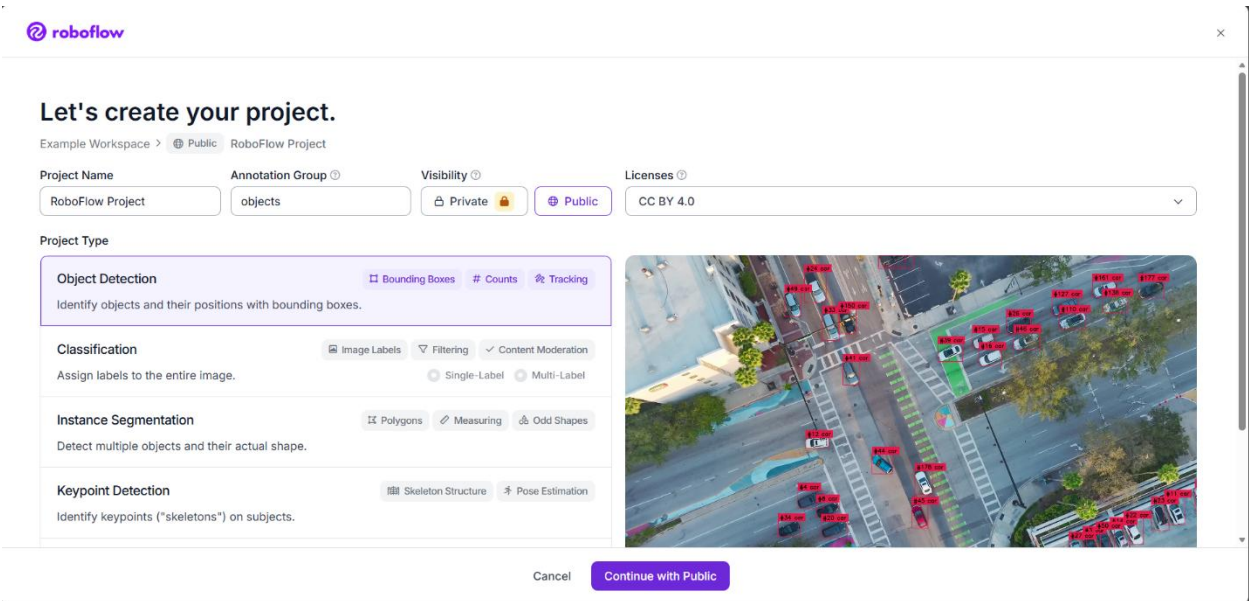


Figure 4. RoboFlow Projects Tab

14. After selecting “New Project”, Figure 5 will appear. Configure the Project parameters accordingly:

- a. *Project Name*: Anything recognizable to RoboCats team. Recommended name is “RoboSub (insert current year) Competition”.
- b. *Annotation Groups*: objects
- c. *Visibility*: Public (Private visibility requires paid subscription).
- d. *Licenses*: CC BY 4.0

e. *Project Type: Object Detection*



The image shows the RoboFlow project creation interface. At the top, the RoboFlow logo is visible. Below it, the heading "Let's create your project." is displayed. The interface includes a breadcrumb trail: "Example Workspace > Public > RoboFlow Project". There are four main sections for configuration: "Project Name" (set to "RoboFlow Project"), "Annotation Group" (set to "objects"), "Visibility" (with "Public" selected over "Private"), and "Licenses" (set to "CC BY 4.0"). The "Project Type" section is expanded, showing four options: "Object Detection" (selected, with sub-options "Bounding Boxes", "Counts", and "Tracking"), "Classification" (with sub-options "Image Labels", "Filtering", and "Content Moderation"), "Instance Segmentation" (with sub-options "Polygons", "Measuring", and "Odd Shapes"), and "Keypoint Detection" (with sub-options "Skeleton Structure" and "Pose Estimation"). To the right of these options is a preview image of a street scene with various objects (cars, pedestrians, etc.) labeled with red bounding boxes and identification numbers. At the bottom of the form, there are two buttons: "Cancel" and "Continue with Public".

Figure 5. Project Options

15. Select “Continue with Public”.
16. RoboFlow constantly updates its vision detection models. After selecting “Continue with Public”, it may prompt you to use its newest model (SAM3, YOLOv38, etc). You will see a “Build with Traditional Model”. Select that button.
17. Your project is created! Your screen should now look like Figure 6:

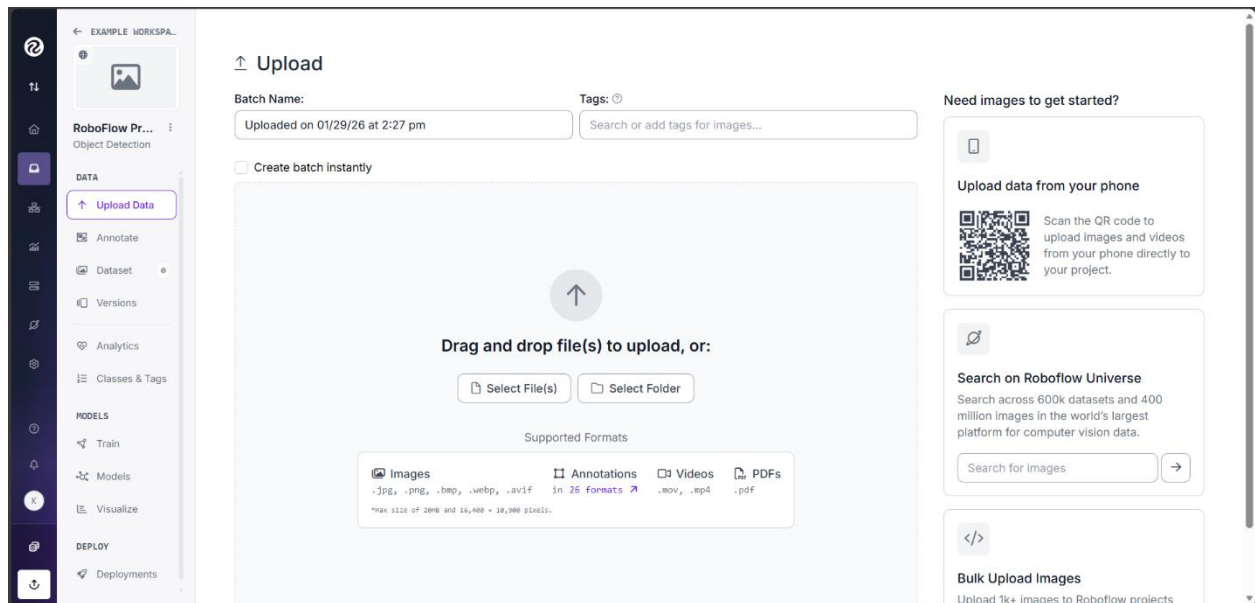


Figure 6. RoboFlow Project Created

18. Your project is now ready to upload images. The setup for both routine model creation and the RoboSub Competition is now complete.

Model Creation – System Setup

System Setup:

This setup is essential for the model creation scripts to work correctly. These steps will cover basic Tempest, Linux and Git usage for someone without any prior experience with these programs. However, I highly recommend using the resources listed in the Troubleshooting section to get more comfortable with these programs.

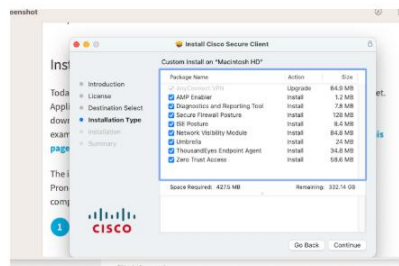
In order to use these scripts, you must have an account with Roboflow, Tempest, and Github. The following steps take you through the sign up process.

Tempest requires you to be on a secure MSU network. On campus, you can use the MSU-Secure Wi-Fi. However, at competition you don't have access to this Wi-Fi so you need to set up a VPN shown in the following steps.

1. If you already have a MSU-Secure VPN configured, you can skip to Step 8.
2. To download the MSU VPN, go to this link:
<https://www.montana.edu/uit/computing/desktop/vpn/>
3. The link will take you to a Montana State page that walks you through how to download the VPN. For backup assistance, a few steps will be briefly outlined here as well. Once at the link, scroll down until you see the image in Figure 1.

Secure Client: Download Instructions

The new VPN requires the Secure Client to connect. After installing it, you will always use the Secure Client app to connect to the VPN. You will not need to go to network settings or use the network connection previously created for the old VPN. Simply open the app and click connect.



Follow the instructions below to download the client and connect to the new VPN.

Desktop/Laptop*

- [Mac](#) (10.9 and newer)
- [Windows](#) (8.1, 10,11)

Figure 7 VPN Download

4. Select the appropriate version to download for your laptop's operating system. Note: this procedure has been written for use with Windows.
5. After going to the link, it is very straightforward and similar to downloading other software. The install wizard will open, and you will be able to install on your device. Figure 2 shows how to access the VPN from a Windows device:

Connect:

1. Open **Cisco Secure Client**.

The app can be found by clicking the small white up arrow in desktop tray and then clicking Cisco Secure Client globe-icon.



The Ready to Connect VPN dialog box opens.

2. Enter **vpn.msu.montana.edu** (if not already entered) and click **Connect**.
3. Log in:
 - a. Choose **Group** appropriate to your role at MSU (in most cases this will be **MSU-Employee-VPN** or **MSU-Student-VPN**)
 - b. Enter your **NetID** and **NetID Password**.
 - c. Click **OK**.

Figure 8 Connecting to VPN

6. You will need to select the correct GROUP. For the user of this procedure, use MSU-Student-VPN. Enter your NetID email (ex: a12b345@student.montana.edu) and password.
7. Figure 3 will show up on your computer now when the Cisco Secure Client icon is selected from the desktop tray. You will be able to connect to it whenever, and wherever.

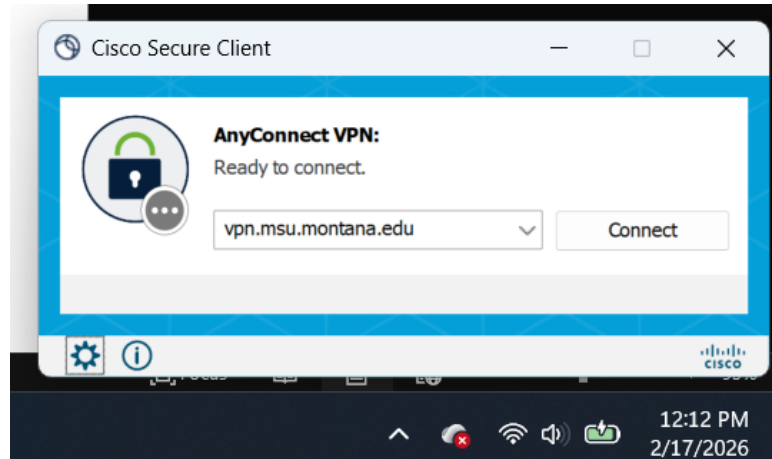


Figure 9 Cisco Secure Client

8. Now, with the VPN step taken care of, it's time to focus on creating a Tempest account. RoboCats have used Tempest to create the model for many years, so it's likely that a member of the team already has a Tempest account. However, this procedure is designed so that any member of RoboCats can follow it. The following steps will go over creating a Tempest account, but if you already have one, you can skip to Step 14.
9. Enter this URL in your computer: <https://www.montana.edu/uit/rci/tempest/>
10. Select the “New User Request” button that is present on this page. Selecting this button will take you to a short survey. This survey will ask you for your name, your email, your affiliation with Montana State (undergrad, grad student, postdoc, etc). Fill out the survey. To use Tempest, your account must be granted access.
11. The survey will ask for the name and email of your sponsor or Lab PI. The current sponsor of RoboCats is Dr. Bradley Whitaker (bradley.whitaker1@montana.edu). You will be asked about the work you would like to do on Tempest. Type in some version of “I am a member of RoboCats, who needs to use Tempest for the RoboSub competition.”
12. Submit your request. Your request will need to be OK'ed by your sponsor (Dr. Whitaker), and then OK'ed by the Research Cyberinfrastructure Team. You can reach them to inquire about the state of your request at 406-994-1777, or at rci-support@montana.edu.
13. The time it takes to get access to Tempest varies. Typically, you can expect it done within 3 business days.

14. Once you have an account, you can again go to this URL:

<https://www.montana.edu/uit/rci/tempest/>

15. Select the “Tempest Web” button. If you have an account, and are either connected to MSU-Secure or using the Cisco Secure VPN, your screen will look like Figure 4.

- a. If you get an error saying “Site can’t be reached” it means you are not currently connected to the VPN, go back to step 6 to connect.

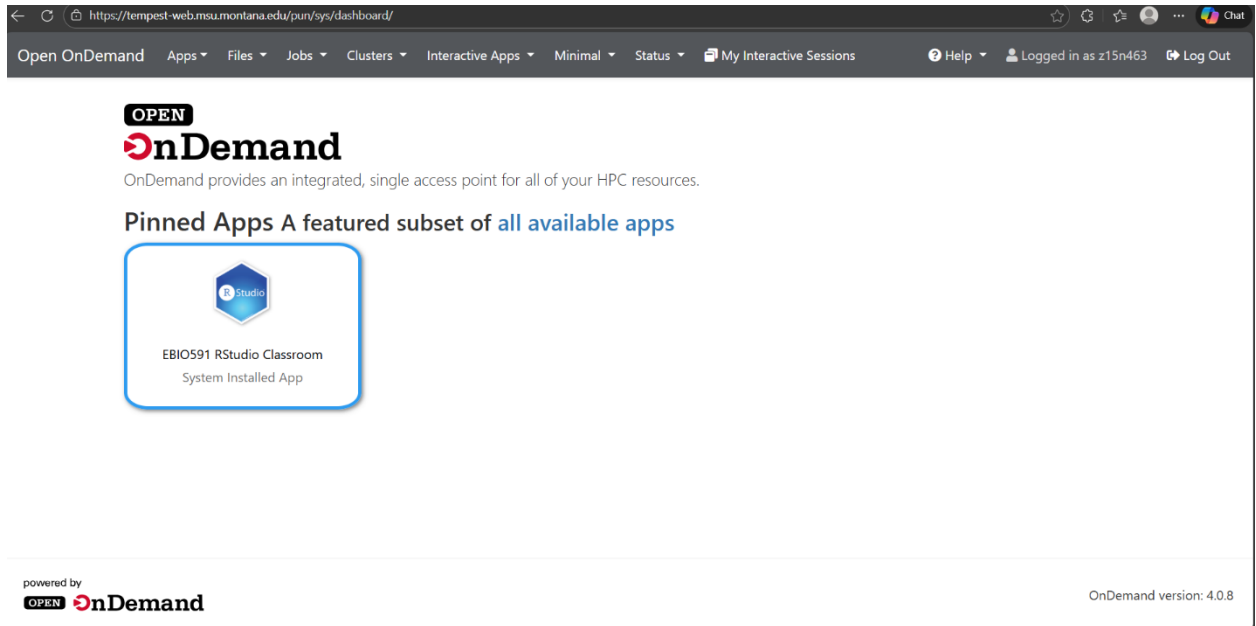


Figure 10 Tempest Home Screen

16. Select the “Files” dropdown menu (top left of Figure 4), and select “Home Directory” from within the drop down options. This will take you to your home directory.

17. To get to the terminal, select the “>_ Open in Terminal” button that appears in your home directory screen. Your screen should now look like Figure 5.

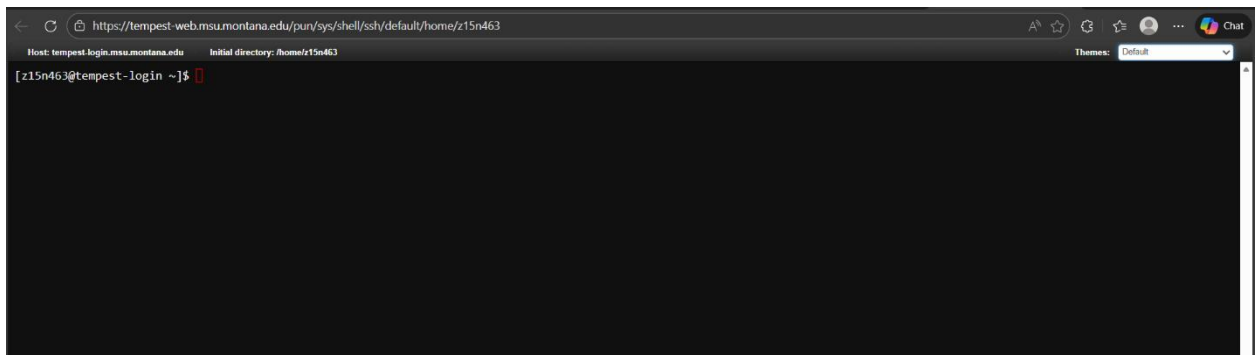


Figure 11 Tempest Terminal

18. Now, you need to setup your `Training` environment on Tempest. You only need to setup the “Training” environment once, and can access it again in later usages. If you have already done this, go to step 23.
19. To create the `Training` environment, you need to run some commands from the terminal to setup the correct software environment on Tempest. Run these commands exactly:

```
cd $HOME
module load Anaconda3/2024.02-1
conda create -n Training python=3.12 -y
source activate Training
```

These commands set up a specific workspace (called an environment) within Tempest, with the correct software tools to train a computer vision model.

20. You should now see “(Training)” to the left of your username in the terminal, like Figure 6.

A terminal window with a black background and white text. The prompt is "(Training) [z15n463@tempest-login ~]\$" followed by a red cursor bar. The text is in a monospaced font.

Figure 12 In the Training Environment

21. Run these commands next, one at a time:

```
pip3 install --upgrade torch torchvision torchaudio
pip install poetry
poetry config virtualenvs.create false
```

These commands install torch dependencies and Poetry. “torch” is used to access and manage the GPUs in Tempest, using CUDA, for the python YOLO model to train on. Poetry is used in the RobosubCV repository to manage all other dependencies.

22. You have now created an environment on Tempest, and installed the correct software needed to move on to the next step. You can use the command `conda deactivate` to exit the environment.
23. Now that the Tempest environment is set up, we need to download the model training software from GitHub. GitHub is a website that stores and shares code. The training software was created by a member of the 2025-26 RoboSub Capstone group, and is hosted on their GitHub page: <https://github.com/kyebuerkle/RobosubCV>. If you already have the repo, go to step 39.

24. You will need a GitHub account for the next parts. If you already have one, you can skip to Step 27. If not, the next steps will walk you through setting up an account.

25. Type in this URL: <https://github.com/>

26. Navigate to the top right corner of the website, and click the “sign up” button. Enter your information to create an account.

27. Now, we need to clone the RobosubCV Repository into Tempest. If you have already done this, go to step 39.

28. Enter the command:

```
git clone git@github.com:kyebuerkle/RobosubCV.git
```

29. If you just setup your GitHub account on Tempest, you will get an error “unable to access repository”. This is because you need to confirm your account with an SSH key in Tempest to be permanently signed in. If you did not get an error, and the repository is cloned, go to step 38. Run this command to generate your ssh key in Tempest :

```
ssh-keygen -t ed25519 -C "your_email@example.com"
```

(Replace the email part of the command with the email you used to sign up for GitHub)

30. In the terminal now, Tempest will prompt you for some things, like “default file” and “passphrase”. Simply hit the **enter key** for every prompt.

31. Now, run the command: `cat ~/.ssh/id_ed25519.pub`

32. That command will display your key. Copy the entire output from the command, including your email.

33. Now you need GitHub to recognize Tempest as one of your devices. Go to github.com in your browser, and go to the SSH Keys: Github → Settings → SSH and GPG keys. The “Settings” tab is located by pressing your username icon in the top right corner of the GitHub website.

34. Your screen should look like Figure 7.

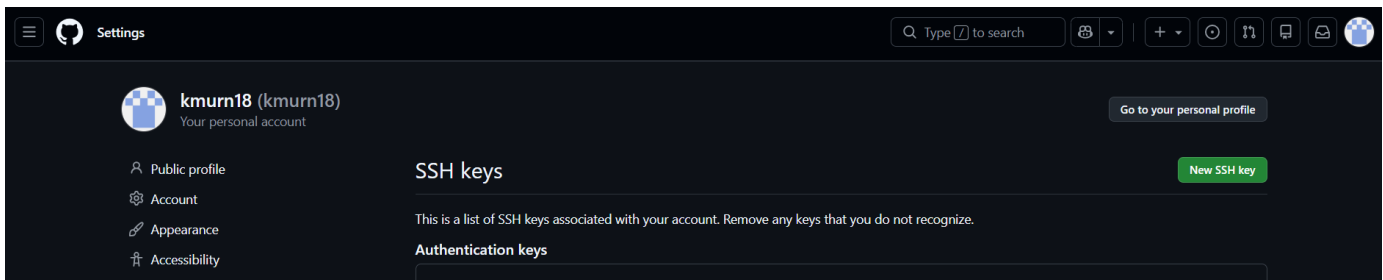


Figure 13 SSH and GPG Keys

35. Select the green “New SSH key” on the right. Give your key a name. The name can be anything, it doesn’t matter. Then paste your key (from step 33 in Tempest) into the main box. It should say “Success!” in the key box.

36. To confirm that GitHub recognizes your account on Tempest, verify with this command:

```
ssh -T git@github.com
```

a message: “Hi [your username]! You've successfully authenticated, but GitHub does not provide shell access.” confirms your account.

37. Time to clone the repository again, run this command:

```
git clone git@github.com:kyebuerkle/RobosubCV.git
```

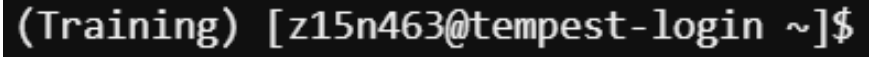
38. Figure 8 shows what the terminal outputs should look like.

```
(Training) [z15n463@tempest-login ~]$ git clone git@github.com:kyebuerkle/RobosubCV.git
Cloning into 'RobosubCV'...
remote: Enumerating objects: 380, done.
remote: Counting objects: 100% (67/67), done.
remote: Compressing objects: 100% (44/44), done.
remote: Total 380 (delta 29), reused 52 (delta 22), pack-reused 313 (from 1)
Receiving objects: 100% (380/380), 2.31 MiB | 4.64 MiB/s, done.
Resolving deltas: 100% (217/217), done.
Updating files: 100% (32/32), done.
```

Figure 14 Cloning Repository

39. To navigate into the repository, run this command (use ‘tab’ to auto-complete after typing a few letters in ‘Robosub’): `cd RobosubCV`

- a. `cd` is a Linux terminal command meaning “change directory” it is used to navigate through folders in your terminal. If you are ever lost just run `cd` to go home.

- b. `ls` is a command meaning “list”, that lists out the files and folders under your current folder. Use this before `cd` to find out which directories to change to.
40. In order to use any scripts in this repository, you need to be in the ‘Training’ environment. (step 19). To activate this environment type:
- ```
source activate Training
```
- you should see “(Training)” to the left of your username in Tempest, shown in Figure 9:
- 
- Figure 15 Training Environment
41. Whenever you access this repository after any changes have been made, you should always pull the changes and download the dependencies (must be in Training environment). It is good practice to do this anytime you use this repo, run the commands:
- ```
git pull
poetry install
```
- This effectively ‘updates’ the repo, or if you just cloned it, ‘initializes’ the repo. (sometimes ‘poetry install’ fails the first time, this is because the dependencies install out of order, type the command again to fix that)
42. The final step in setting up this repository is to select your settings. The `setup_settings.py` script will bring you through the proper steps to: 1. Login to Roboflow, 2. Select output folders, 3. Edit the augmentations for training.
43. Run the command, and follow the steps it goes through:
- ```
poetry run python src/model_training/setup_settings.py
```
- a. Press Enter for every prompt to use the default settings.
- b. The ‘`setup_settings.py`’ script is used to configure the ‘`configuration.json`’ file used in the model training scripts. Use the ‘`-h`’ option, following the command, to see all the options. Use ‘`-p`’ to print out the current settings.
44. That is all the basic setup you need to start training a YOLOv8 model!
- If you want to do more complex setup, or understand the repository a bit better, read through the README.md documents in each of the folders. (use commands like: `cat README.md` or `nano README.md` to read the files. It is easier to open the [RobosubCV Github](#) on a browser and scroll through them).

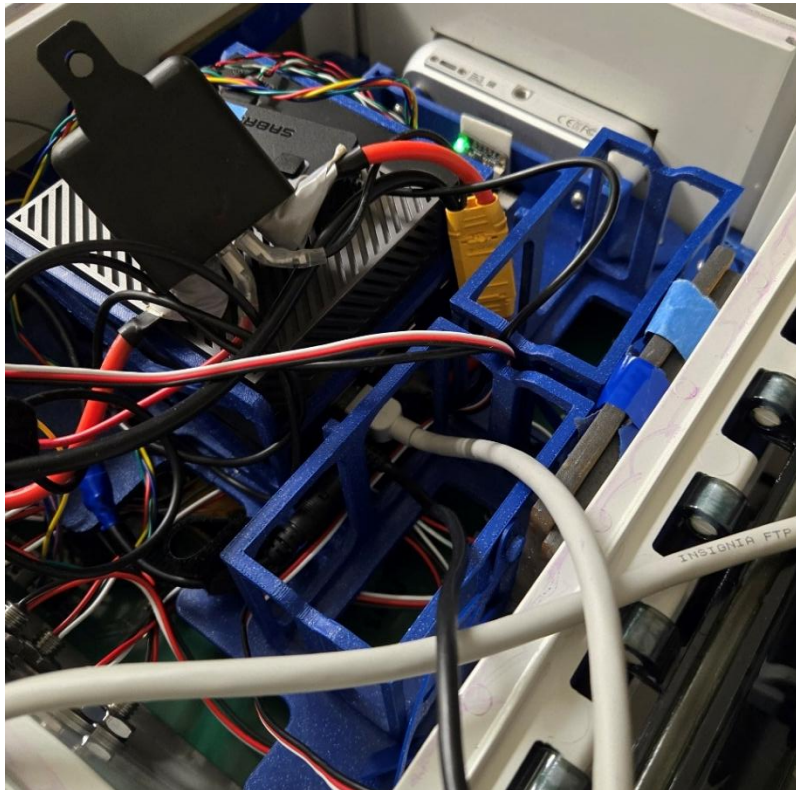
# Model Upload – System Setup

## System Setup:

This section covers how to set up a connection between your personal laptop and the Jetson Nano computer onboard the submarine. This connection is used to transfer the trained computer vision model from your laptop to the submarine. This setup only needs to be done once, before competition.

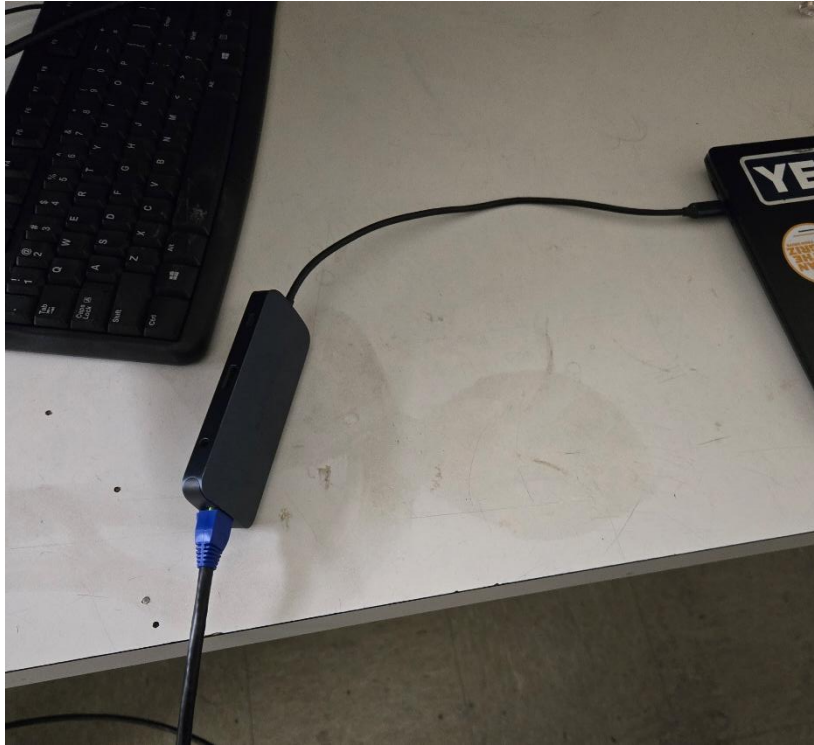
**Prerequisites:** You must have already completed the Model Creation section and have a trained YOLOv8 model file (.pt file) downloaded to your laptop.

1. Locate the trained YOLOv8 model file and move it to your Downloads folder. The file will be named something like "yolov8m.pt" or similar.
  2. Connect the Jetson Nano computer to power using a USB-C cable, as shown in Figure 16
- Jetson Nano with Power Plugged in



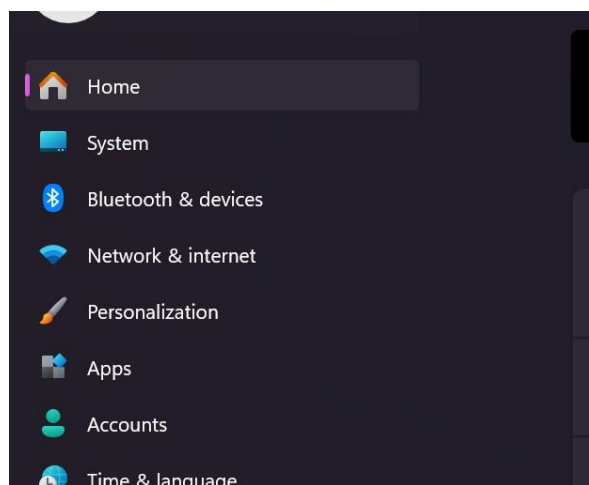
*Figure 16 Jetson Nano with Power Plugged in*

3. Connect an Ethernet cable between the submarine's Jetson Nano and your personal laptop, as shown in Figure 2. This is a direct connection - it does not go through a router or switch.



*Figure 17 Ethernet connection with Robosub and Personal Laptop*

4. On your laptop, open Settings and navigate to "Network & Internet," as shown in Figure 3



*Figure 3 'Network & Internet' in Settings of Windows 11*

5. Navigate to 'Ethernet'

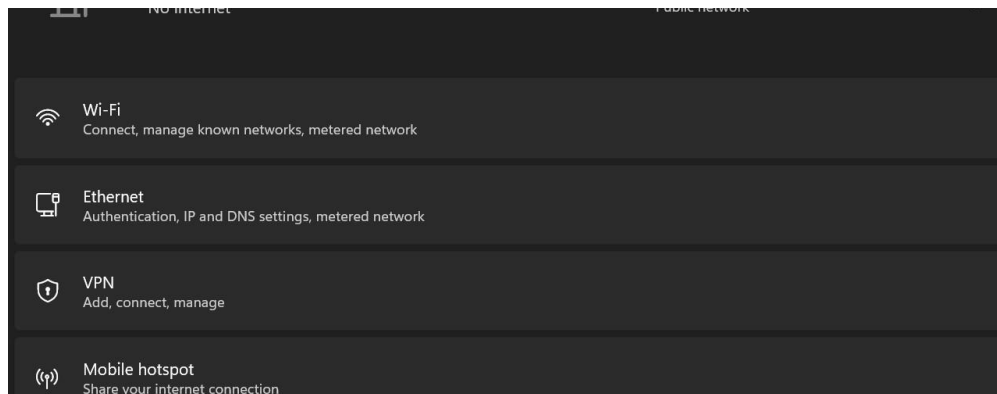


Figure 18 Ethernet settings within Network and Internet

6. Click on edit along 'IP Assignment'

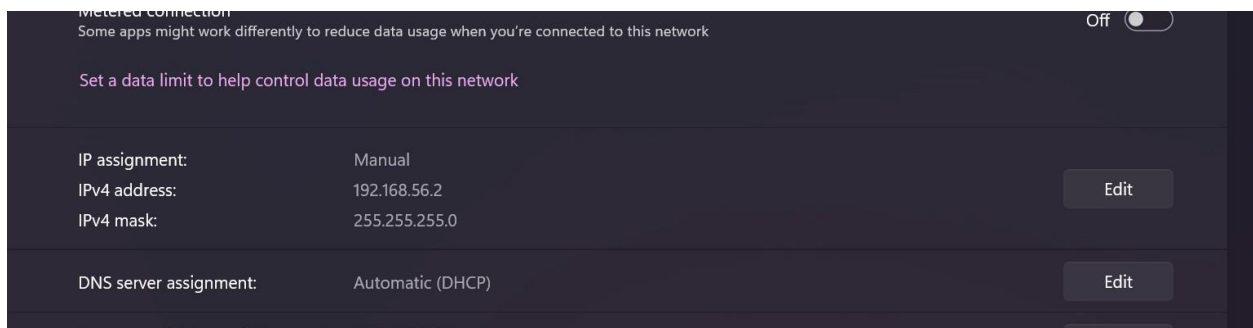
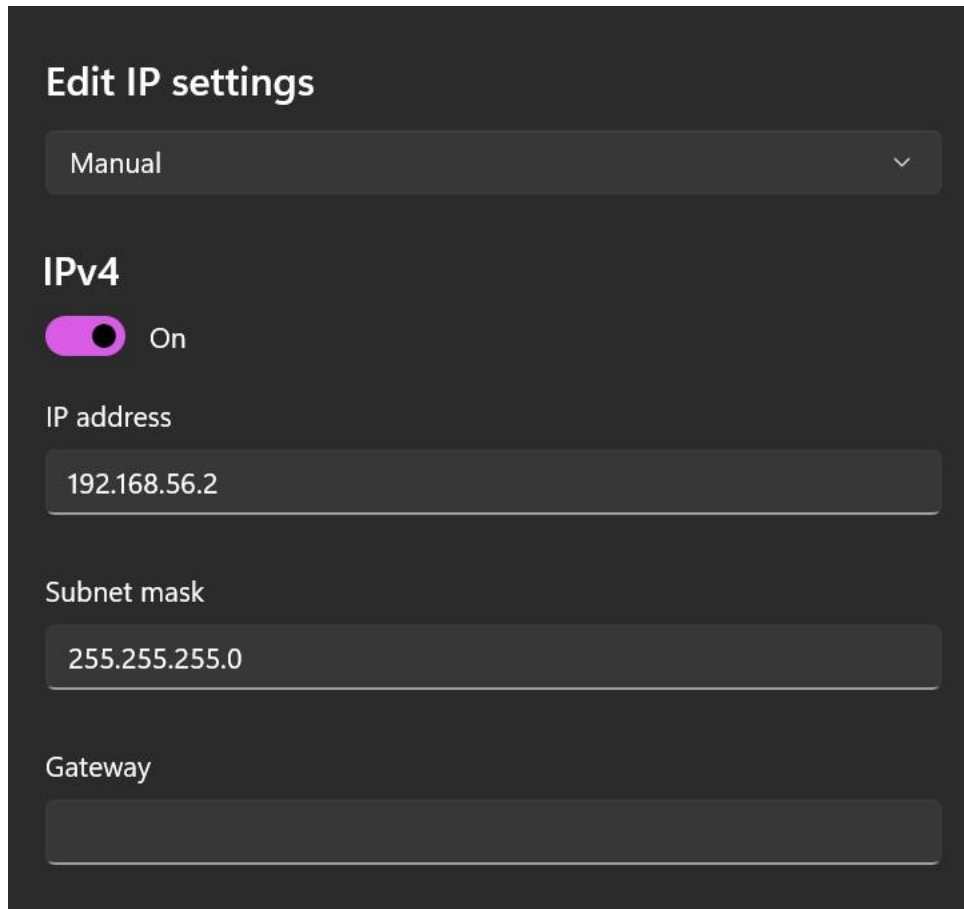


Figure 19 Section to change IP assignment

7. Toggle IPv4 to "Manual" and enter the following values exactly as shown in Figure 6. These settings allow your laptop to communicate directly with the Jetson Nano.



The image shows a dark-themed window titled "Edit IP settings". At the top, there is a dropdown menu currently set to "Manual". Below this, the "IPv4" section is active, indicated by a blue toggle switch labeled "On". Under the "IPv4" section, there are three input fields: "IP address" containing "192.168.56.2", "Subnet mask" containing "255.255.255.0", and "Gateway" which is currently empty.

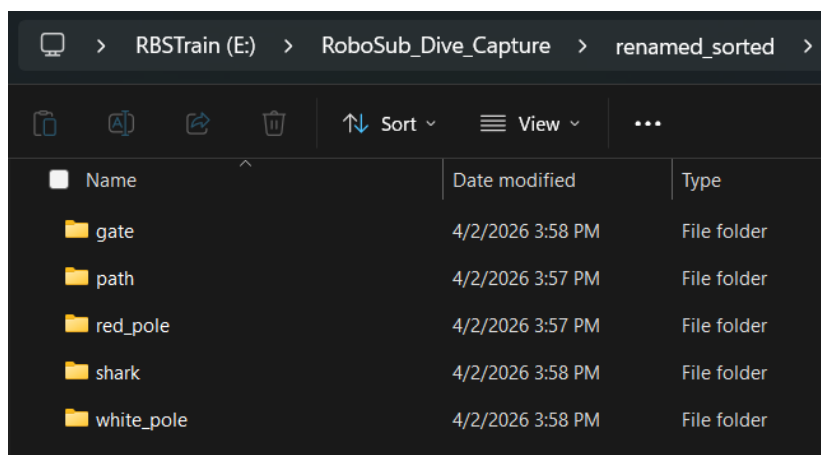
*Figure 20 IP address and Subnet mask*

8. Click "Save." Your laptop is now configured to communicate with the submarine's computer.

## Data Collection – Competition Workflow

This section walks through the steps of collecting raw video footage of competition objects using a GoPro camera. The diver will record video of each competition object from multiple angles and distances. These videos will later be processed into individual images that are used to train the computer vision model. Before entering the water, verify that the camera is properly prepared.

1. Prepare the camera for data capture. Ensure that the camera has greater than 50% charge, and that the MicroSD card has more than 2 hours of footage storage on it.
2. Set the GoPro to video mode, resolution 1920x1080p, frame rate 30fps, and linear FOV mode. Ensure that the camera lens is clean, and if not use a microfiber cloth to clean it.
3. Remove the MicroSD card from the GoPro and connect it to your laptop.
4. Open the file explorer and navigate to the MicroSD card, **RBSTrain (E:)**
5. Open **E:\RoboSub\_Dive\_Capture**
6. Open **E:\RoboSub\_Dive\_Capture\renamed\_sorted**
7. Create a folder for each class of object present in the competition pool. For example, if there is a path, red pole, white pole, shark, and gate the folder should look like the following figure. These folders will be used to sort the captured videos. Folder and file names should not have any spaces in them, as they will be parsed by a script, and spaces in folder and file names can cause a parsing error.



8. The seven different section folders will be used at a later point for helping to label the frames.
9. Close the file explorer, eject the MicroSD card, remove it from your MicroSD card reader, and reinsert it into the GoPro. You can eject the external drive by clicking the little up carrot symbol on the bottom of your screen in the taskbar near the date and time, clicking the flash drive symbol, and clicking eject **RBSTrain (E:)**.
10. Insert the GoPro into its case and turn it on.



11. Pick an order in which to capture the objects present in the competition environment. If there are multiple objects in one class, capture all of them one after another before capturing other objects in the competition environment. Do this so the later sorting step will be easier. Then move onto the next nearest object class, and repeat the order.
12. Enter the water.
13. Approach the object and align yourself to it relative to one of the four cardinal directions relative to the pool walls. Basically, have two walls perpendicular to yourself in front and behind you, and the side walls parallel to your field of vision. This is done so it will be easier to tell when a full 360° circle of the object has been completed. Approach the object to a distance of 1.5 meters with the camera at approximately the same depth as the object.
14. Start the video recording.
15. Start to swim, walk, or tread water around the object, based on pool depth and object depth in the pool. Encircle the object in a clockwise manner. Keep movement speed to less than 0.5 meters per second to minimize motion blur.
16. Keep the object at least 75% present in the frame at all times and ensure that the object does not ever completely leave the cameras field of view. If the object completely leaves the field of view, for every second it is out of frame add two seconds of video time with the object in frame.
17. Once the 1.5 meter encirclement of the object is complete, stop recording and move back to 4 meters from the object.
18. Repeat steps 14, 15, and 16 at a distance of 4 meters from the object.
19. Identify what object or class of objects is next in the predetermined order of capturing the objects in the competition environment.
20. Repeat steps 13 through 18 on the selected object.
21. Repeat steps 13 through 20 until all classes of objects present in the competition environment have been captured.
22. Once all objects in the predetermined order of objects to be captured have been captured on the GoPro, exit the pool.
23. Remove the GoPro from its housing.
24. Ensure that the GoPro is dry before opening up the watertight compartments on it, to make sure that water does not get inside and fry your GoPro.
25. Remove the MicroSD card from the GoPro and plug it into a microSD card adapter connected to a laptop.
26. Open the file explorer and navigate to the folder on the SD card where the GoPro stores captured images and videos, **E:\DCIM\100GOPRO**.
27. Select the View option on the top of the screen under the address bar at the top of the screen.
28. Select either large or extra large icons. This makes it so you can see a thumbnail of the video, and makes it much easier to see what object each video captured.

29. Select all of the .LRV and .THM files present in **E:\DCIM\100GOPRO**. These files are low resolution video and thumbnail files used for previewing files on the GoPro and the Quik app. Delete all of these files. They are not used in this procedure, and simply take up space on your computer screen.
30. Cut all videos captured in the pool from **E:\DCIM\100GOPRO** to **E:\RoboSub\_Dive\_Capture\raw\_videos**.
31. Open up another tab in the file explorer and navigate to **E:\RoboSub\_Dive\_Capture\renamed\_sorted**.
32. Copy all videos captured of an object to its respective folder in **E:\RoboSub\_Dive\_Capture\renamed\_sorted**, the folders that were created in step 7. Do not worry about renaming the video files, as the data ingestion script will automatically rename the video files based on the name of the folder that they are in.
33. Open up Windows PowerShell.
34. Set your current working directory to the scripts folder on the MicroSD card with the command **cd E:\Scripts\_RBSTrain\**.
35. Run the data ingestion script using the command **python .\data\_ingestion.py**.
36. The script will prompt you for a dataset root directory. Enter **E:\RoboSub\_Dive\_Capture**. The reason it does not default to this is in case there are other copies of this dive capture folder for different dives. If you use a different dive capture folder, just enter the root directory for that folder.
37. The script menu will come up.

```
Dataset root directory: E:\RoboSub_Dive_Capture

Dataset root : E:\RoboSub_Dive_Capture
renamed_sorted : E:\RoboSub_Dive_Capture\renamed_sorted
640x480_frames : E:\RoboSub_Dive_Capture\640x480_frames
rejected_frames : E:\RoboSub_Dive_Capture\rejected_frames
pruned_500_all : E:\RoboSub_Dive_Capture\all_objects

=== RoboSub Dataset Menu ===
 1) Rename videos
 2) Extract frames
 3) Rename and extract frames
 4) Remove blurry frames (move to rejected_frames)
 5) Restore rejected frames
 6) Cap frames to 500 per class
 7) Delete all extracted frames
 8) Quit
Choose [1-8]: |
```

38. Select option 3 to rename all of the captured video files and extract frames from the videos. The frames will be scaled down to the resolution used on the RoboSub, 640x480p, and saved in the **640x480\_frames** folder in a subfolder created by the script for each object..
39. Once the script has finished renaming the videos and extracting frames, press enter to return to the main menu.

40. Select option 4 to remove blurry frames from the datasets. All removed frames will be moved to the **rejected\_frames** folder, and will not be deleted. This is done so they can later be restored if desired, an action performed by option 5, restore rejected frames.
41. After the script has finished removing blurry frames, press enter to return to the main menu.
42. Optionally, select option 7 to remove near-duplicate frames using perceptual hashing (pHash). This action also includes a class-selection prompt and asks for a pHash Hamming threshold, with a default threshold of 4. Frames identified as near-duplicates are moved to rejected\_frames and are not deleted.
43. Select option 6 to cap the number of frames per class to 500 frames. This action includes a class-selection prompt. For each selected class, the script subtracts 500 from the total number of frames to find the number of excess frames E. It then divides the ordered frames into E consecutive groups and removes the blurriest frame from each group. This leaves 500 frames while reducing neighboring removals and helping preserve frame diversity.
44. After blurry-frame removal, optional pHash near-duplicate removal, and frame capping are complete, select option 9 to build the all\_objects folder. This copies all currently kept 640x480 frames into all\_objects using class-prefixed filenames. Because all\_objects is not rebuilt automatically after pruning, this step should be run after the cleanup steps and again after any later restore or delete action that changes the kept frames.
45. Proceed to the labeling procedure. If a train, validation, and test split is needed later, use option 10 after the dataset cleanup is complete.

For actions that operate on object classes, the script will display a class-selection prompt before running the action. The prompt asks whether the action should be run on all classes, only selected classes, or all classes except selected classes. If the user chooses to include or skip specific classes, the script will display a numbered list of available class folders and prompt the user to enter the class numbers separated by commas. This allows the user to apply an action to the entire dataset or only to the relevant object classes without affecting the others. For example, if the blur filter removed far too many of frames for one class, the user can select the option to restore the blurred frames and redo the blur filter and choose to leave out the one class that had too many frames removed.

If you have any trouble completing this procedure section, please refer to the Data Collection Troubleshooting section at the end of the procedure.

**This concludes the Data Collection Procedure Section.**

## Data Preparation - Competition Workflow

This section will walk you through the steps you will take at competition. This section starts off where the **Data Preparation System Setup** section ended, with a RoboFlow account, workspace, and project already created. This section will detail how to take the images from the previous **Data Collection** section, manually label some of the images, create a model on RoboFlow, label the rest of the dataset, and then get the labeled images ready to export to the next subsystems.

1. Before we upload any files to the project, we need to update our file systems to make future processes easier.
2. Remember, the images from Data Collection are stored on an SD card, from the GoPro that was used to take pictures. The SD card should be plugged into your computer now. Open “File Explorer” on your laptop. In the left sidebar, locate and click on the SD card (usually labeled 'Removable Disk' or the camera model name).
3. The SD card should have 1 folder stored on it, with a ton of images in that folder (around 3000 from competition diver). We need to create 7 different folders on the SD card, to store the images in.
4. You are probably asking “Why?” to step 3. The reason is, we need to manually label 175 images to create a model that can label the rest of the batch. We do NOT want to label only the first 175 images in the dataset. The first images might not be representative of what the entire dataset looked like. It is much more prudent to manually label some images from the ENTIRE dataset, for diversity in training images. Thus, we will split the total number of images into 7 different sections, containing equal number of images, and label 25 images from each section. Make sense? Clear as mud?
5. Create 7 different folders on the SD card. Your screen, on File Explorer, should look like Figure 1.

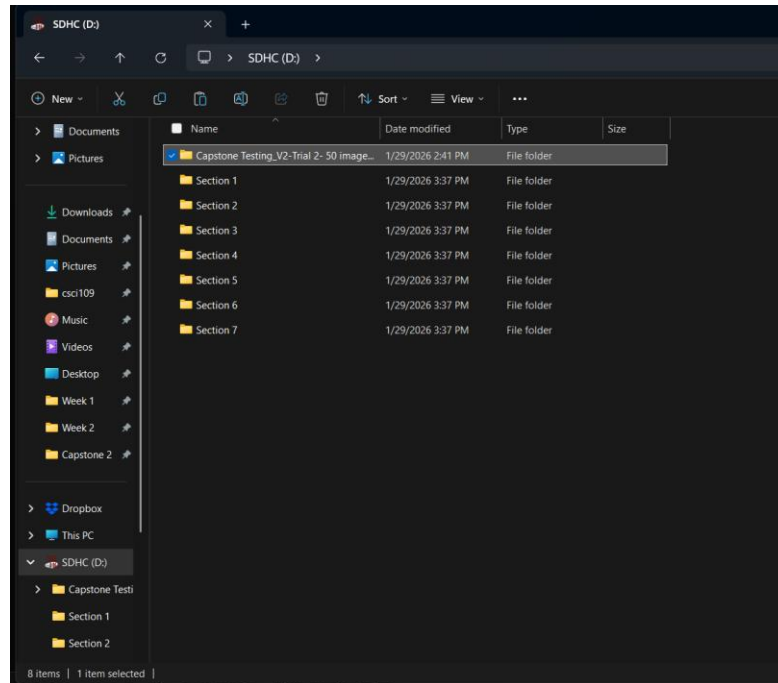


Figure 1. Folders on SD Card

6. The next step is split the images (present in the main folder) into the Sections folder. The amount of images that go in each section folder depends on how many images there are total. If there are 3000 images, then there should be 428 images in each section folder ( $3000/7 = 428.57$ , need a whole number of images).
7. Go into the main folder and select the number of images that should be in the first section. Figure 2 shows me doing this on my screen, with 50 images total in the file (so I am selecting 7 images to move to Section 1).

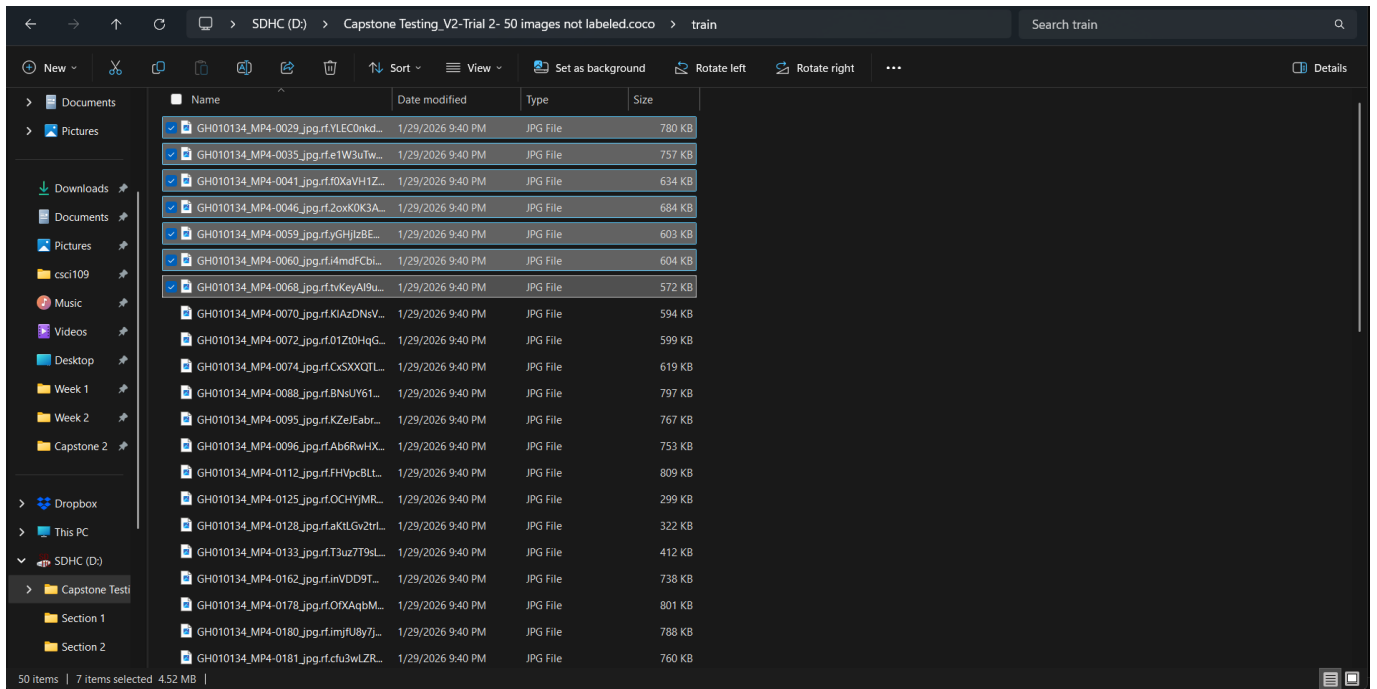


Figure 2. Selecting Images to Move

8. Cut and paste these images into the Section 1 folder. Repeat this process until the images are sorted in the Section folders.
9. At this point, you should have the appropriate amount of images in each folder section. We are now ready to upload these to Roboflow.
10. Open Roboflow on your computer, and navigate to the “Projects” tab on the left hand side. Your screen should look like Figure 3. Open the project.

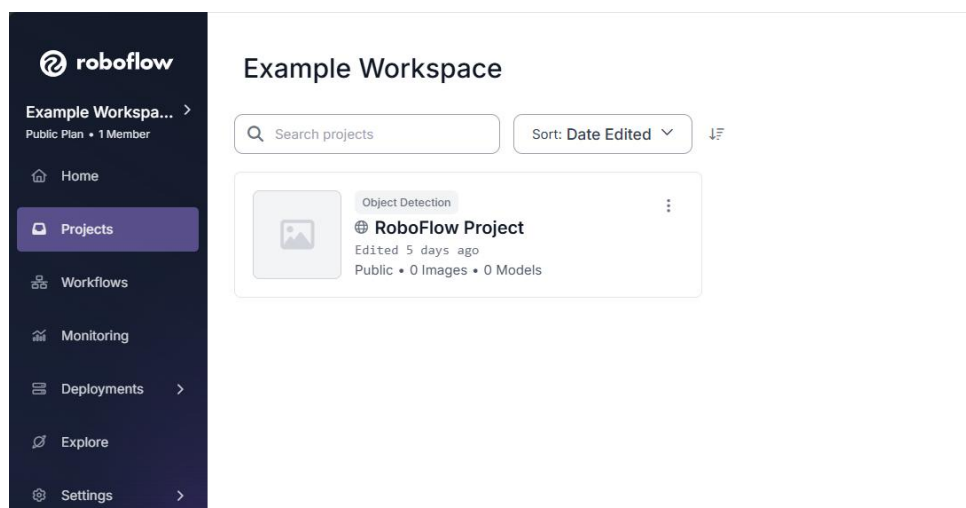


Figure 3 RoboFlow Project Tab

11. Once opening the project, you will automatically be directed to the “Upload Data” tab. To upload images to the project, click “Select File(s)”. **Don’t click “Select Folder”.**
12. After clicking “Select File(s)”, you will need to browse your computer to the SD card. Open Section 1 only, and type “ctrl+a” to select all the images in the Section 1 folder only. It is crucial that we upload images **only 1 section at a time**. This is how we will keep the images separate in RoboFlow.
13. After selecting all the Section 1 folder images, press enter and upload them to RoboFlow. Your screen should look like Figure 4.

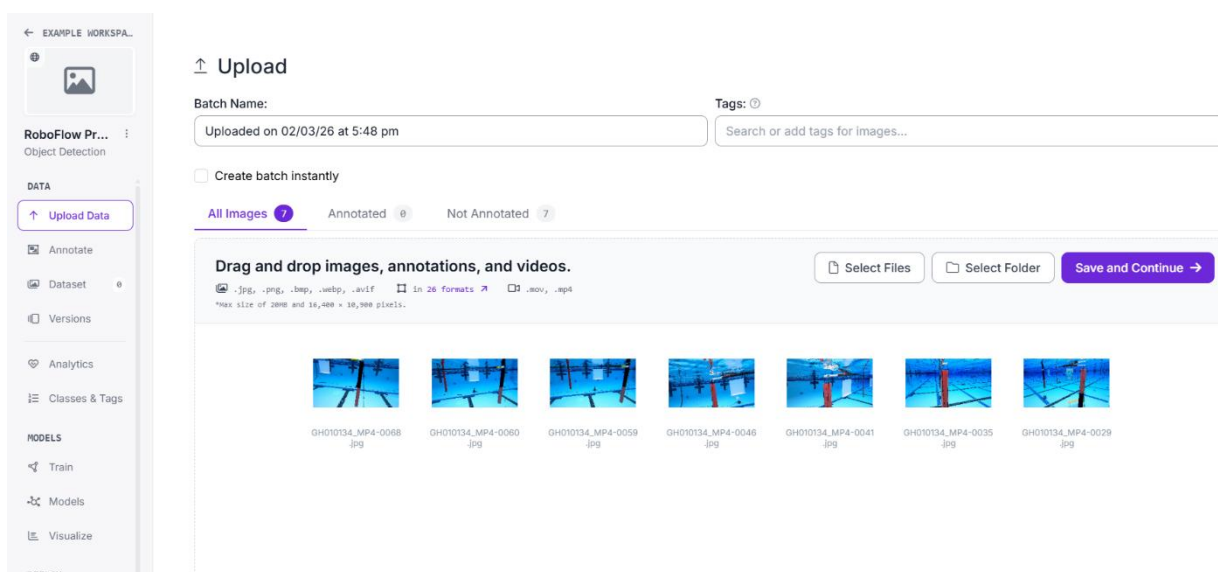


Figure 4 Section 1 Images Uploaded

14. Under “Batch Name:”, rename this batch “Section 1 Images”. Obviously, this gives this section a differentiating name. Select the “Save and Continue” button to put these images into the project. Remember, at this point, we only want the images from the Section 1 folder.
15. After uploading, RoboFlow will automatically take you to the “Annotate” Tab. Your screen will look like Figure 5.



Figure 5 Annotate Tab

16. Select the “Rename” button in the top right corner, and name this batch “Section 1 Images”.  
In Figure 5, there is only 7 images because I’m only using 50 images total. As you follow this procedure, you will have many, many more images in each batch.
17. Go back to the “Upload Data” tab. If you want, you can then click on the “Annotate” tab again, and will see how the images you just uploaded are in their own batch. Upload Section 2, and repeat steps 14-16 to upload all 7 sections.
18. As you complete Step 17, **you must remember to rename each batch**. If you don’t give each batch a unique name (“Section 1”, “Section 2”...), then the images you upload will go into the same batch automatically.
19. If you click on the “Annotate” tab, your screen **MUST** look like Figure 6.



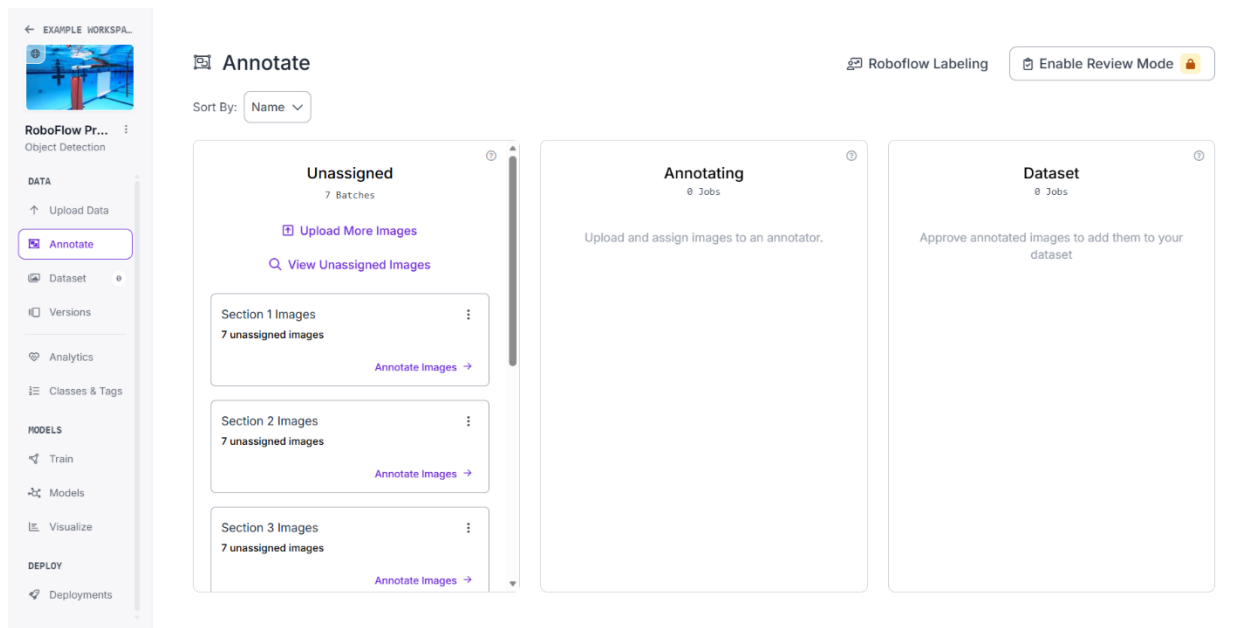


Figure 6 Uploaded Batches

20. At this point, you should have 7 different batches, each with ~400-500 images in them. It is now time to begin labeling them. Remember, we only need to label 175 images manually. The next step, after manual labeling, is to train a model on RoboFlow to label the rest of the images. In each of the 7 sections (batches), you and your team will label 25 images.

21. Under the “Annotations” tab (as seen in Figure 6), select “Annotate Images” for any of the batches. For simplicity, you can start with Section 1. Your screen will then look Figure 7.

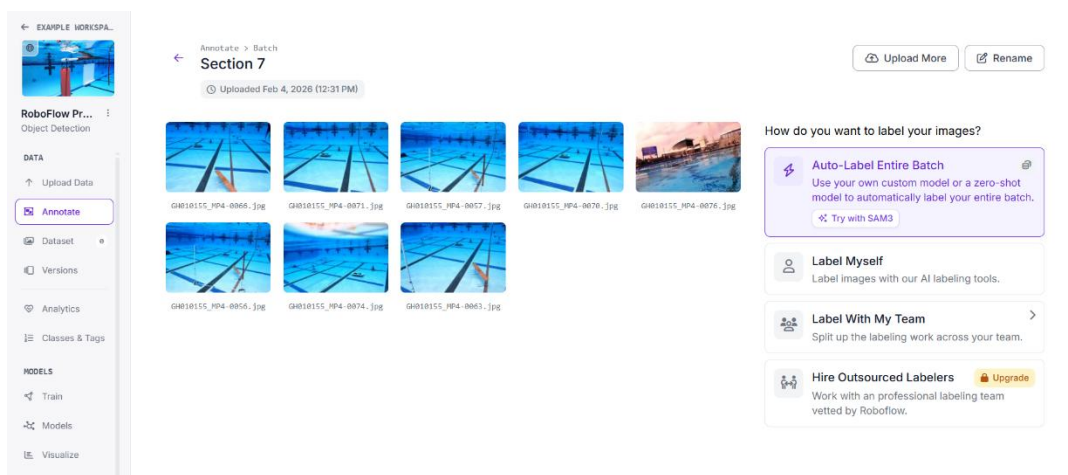


Figure 7 Labeling Options

22. Select “Label Myself”. This will take you to a screen that simply shows you which images are labeled and which are not. At this point, everything will be unlabeled (or unannotated). Select “Start Annotating” from the top of the screen.

23. Your screen will now look like Figure 8.

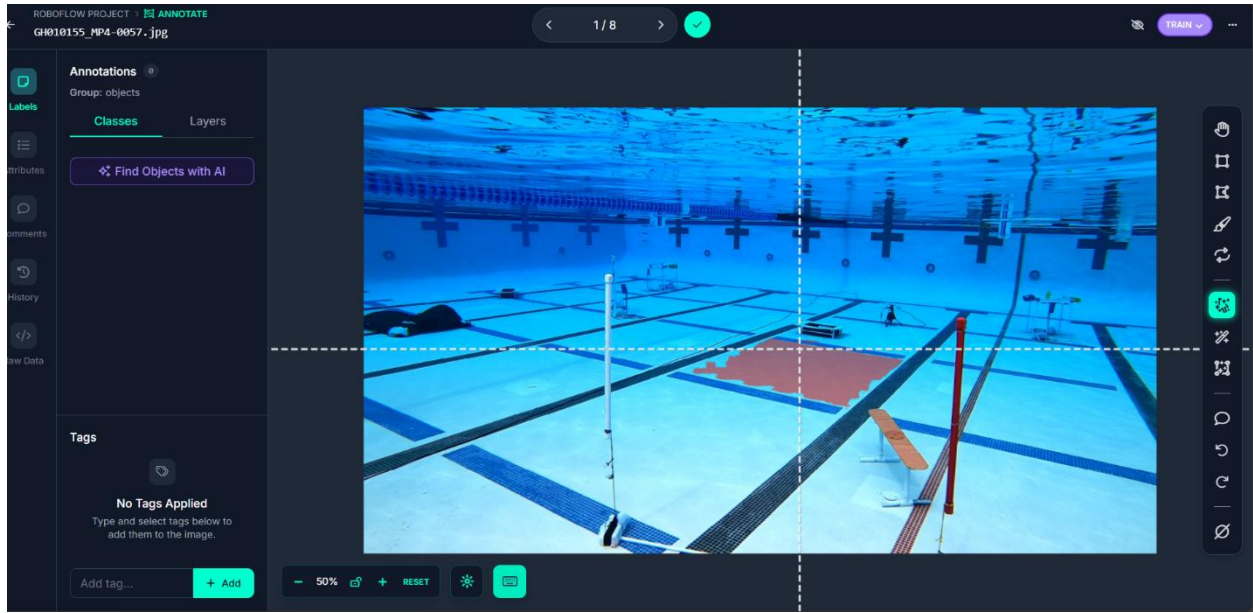


Figure 8 Annotation Screen

24. On the right-side tab, the tool automatically selected is the “Smart Select” (that is what is causing the shade of red where the dotted white crosshairs are in Figure 8). We don’t want that tool. Navigate to the second-from-the-top icon and select the “Bounding Box Tool”.

25. We can now start labeling the images! In Figure 8, the objects we are interested in are the red and white PVC pipes, and the path on the pool floor. Your objects will be different in your images, obviously. Draw a bounding box around your interested image, and a pop-up box will appear, asking you to label what class of object that is. Figure 9 shows an example of this, with me labeling a white pole.

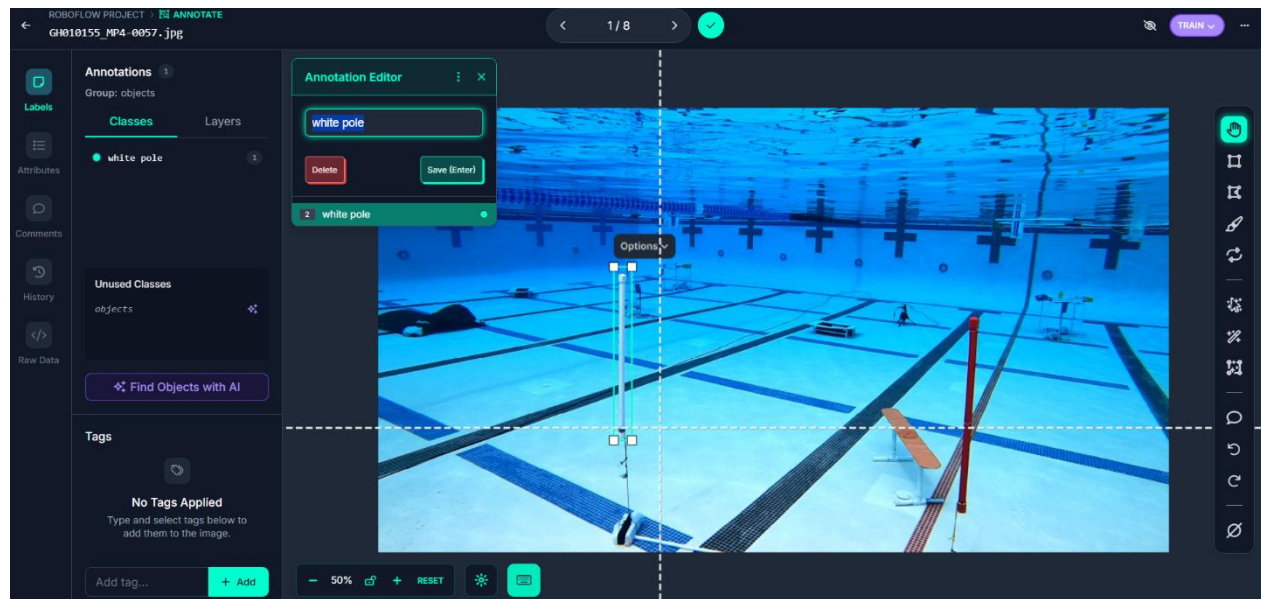


Figure 9 Classes Box

26. Insert a relevant class name. The objects truly are different each year, so I do not know what you will be labeling. Hit the enter key on your keyboard, or select “Save (enter)” with your mouse to save this Class name.
27. Label the rest of the objects in your image, and give them the appropriate Class name. These class names will then be saved to the entire project, meaning that when you start labeling Section 2, Section 3, etc, you will not need to retype the Class names. They will be saved already, and you will select them with your mouse.
28. It can be confusing as to what to label in each image. You may be tempted to label everything, including faint and small background objects. It is my recommendation that you do NOT do this. Simply label the objects that are in the foreground of each image. There is no defined criteria as to what to label, but using your best judgement and more importantly, consistency across all your labeled images is best practice
29. Get to labeling! Remember, we only need 25 images from each section. Your section might have 400-500 images in it. The purpose of dividing up the images into sections was to get 175 *diverse* images, to train a model on. As you label images, they will automatically sort themselves out of the unlabeled images, as I will show in a few steps. You can select the back arrow at any time to exit the annotation screen, and RoboFlow will take you back to the project screen, where you can check how many images you have labeled.
30. Label 25 images in your section. Your screen will now look like Figure 10.

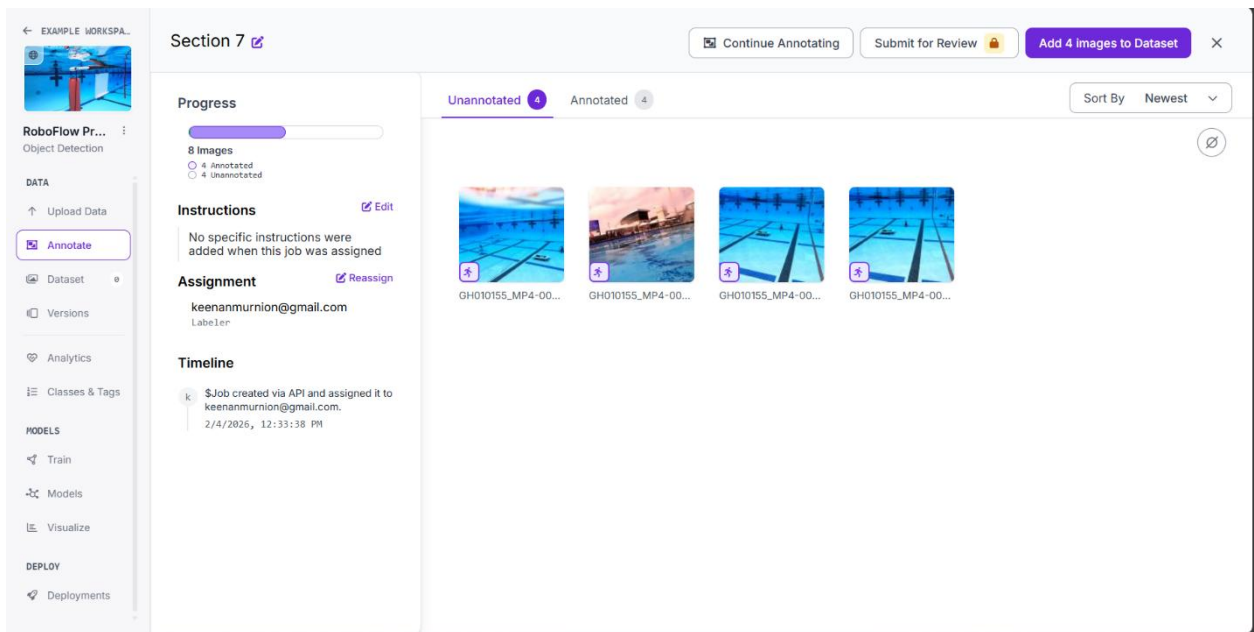


Figure 10 Annotated Images

31. In Figure 10, I have only labeled 4 of my 7 images. This is because 3 of the images had no objects in them! This is just a feature of the images I am using for this example. In practice, by the time you get to this step, you need to have 25 images labeled. Select the “Add 25 Images to Dataset” button in the top right corner, as shown in Figure 10.
32. The box shown in Figure 11 will pop up, asking you how you want to split your images between Valid, Test, and Train. This doesn’t matter at all right now. Leave Method as “Use Existing Values” and select “Add 25 Images” (my button says “Add 4 Images”).

Labeled Images to Add
4

Remaining Batch Images
4

Your remaining images will be split into a new batch.

Method
What's Train, Valid, Test?

Use Existing Values

Distribution
Images with an existing split (4)

Train: 4 images
Valid: 0 images
Test: 0 images

Submit for Review

Add 4 Images

Figure 11 Adding Images to Dataset

33. If you return to the “Annotate” tab, your screen will look like Figure 12. Under the Dataset Column, you will see the images we just uploaded. Under the Annotating Column, you will see the images we did not label (3 images in my example, will be hundreds of images when you do it). At this point, select the 3 dots in the right hand corner of the batch under the Annotating Column, and select “Move to Unassigned”.

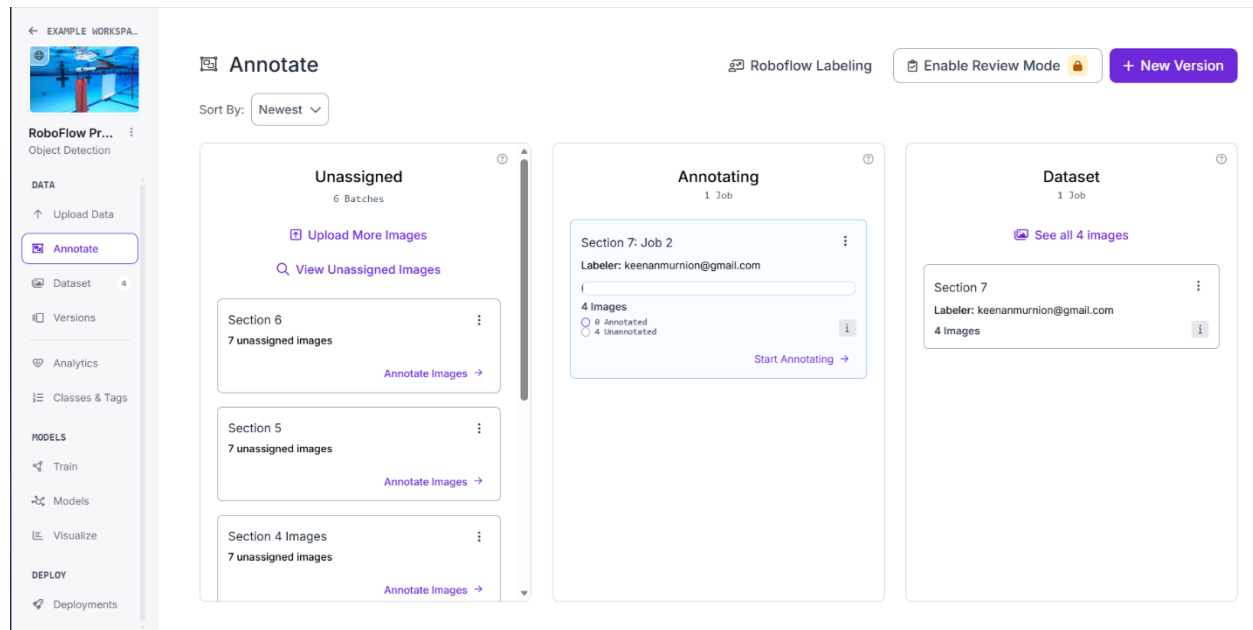


Figure 12 One Section Completed

34. Give the unassigned images a descriptive name, to avoid getting confused. Figure 13 is an example.

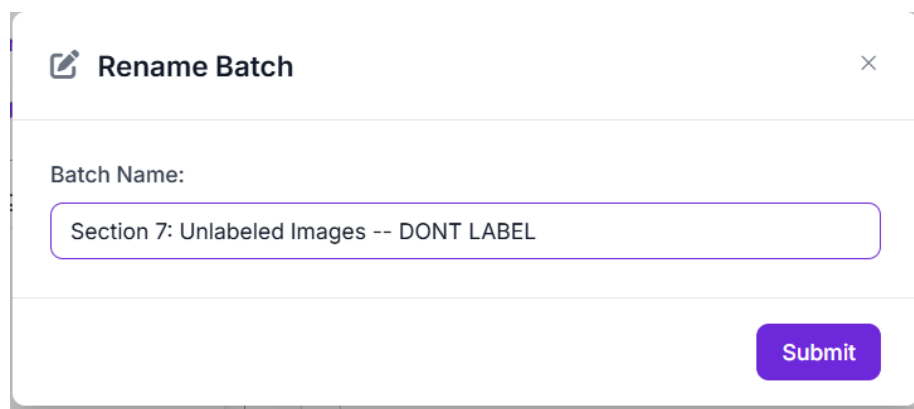


Figure 13 Unlabeled Batch Name

35. Repeat Steps 30-34 for all 7 Sections. Multiple teammates can label concurrently, meaning they can work on different sections while you work. It takes me about 20-30 seconds to label an image, so it should take about 10 minutes to label the 25 images per section. If 4 people are working at the same time, 175 images should be able to label in around 1 hour.
36. When you finish labeling the 175 images, across 7 different sections, your “Annotate” tab should now look Figure 14. The similarities between Figure 20 and your screen will be the 7 sections under the Dataset Column. You can see the “See all 23 images” text in the Dataset column. This is because I only had 23 images with objects in them in this example. You will **need** to have 175 images in your Dataset.

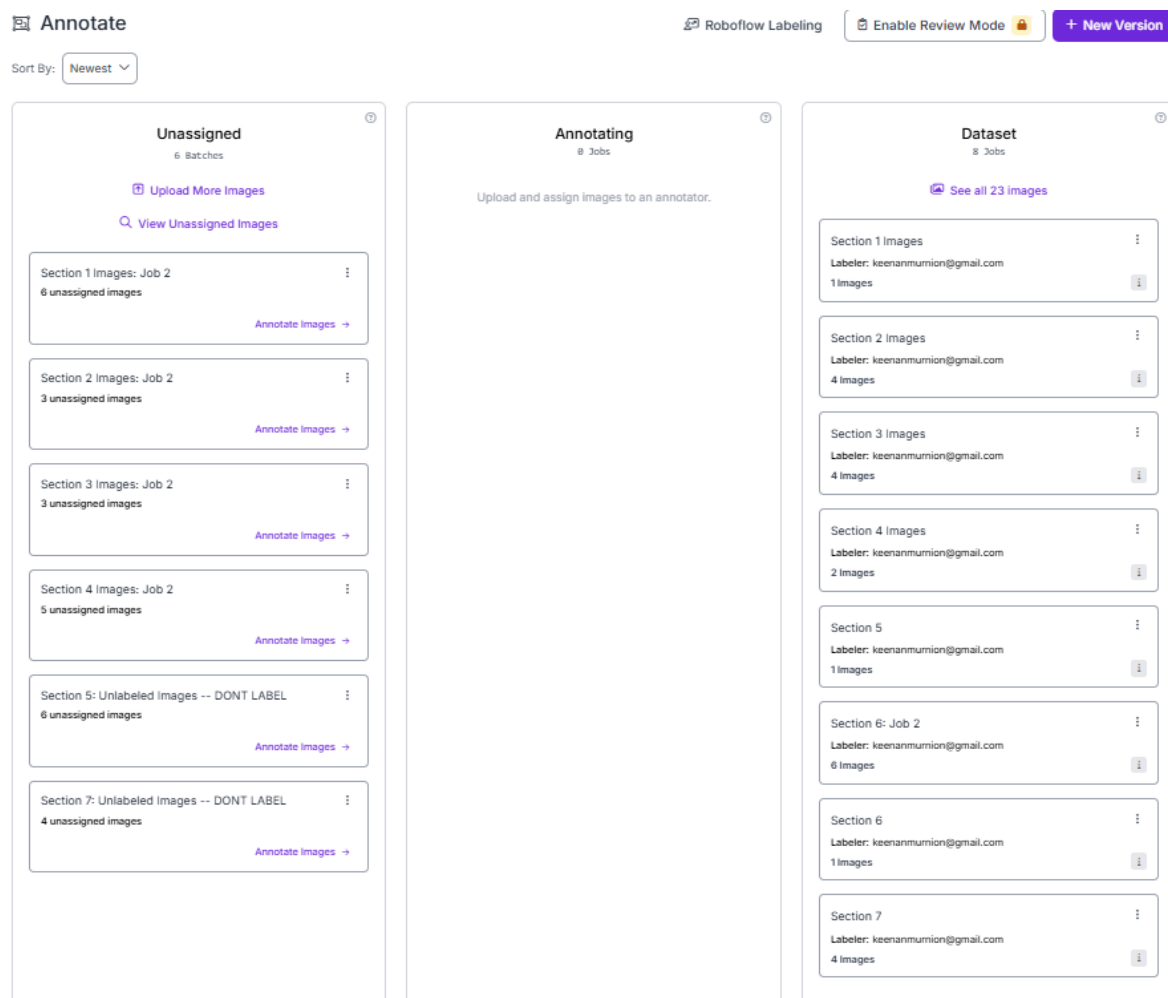


Figure 14 Images Labeled



37. Now, it is time to train a model on Roboflow, using the 175 images you have labeled. Navigate to the “Dataset” tab on the left side bar. Figure 15 is a screenshot of what it should look like.

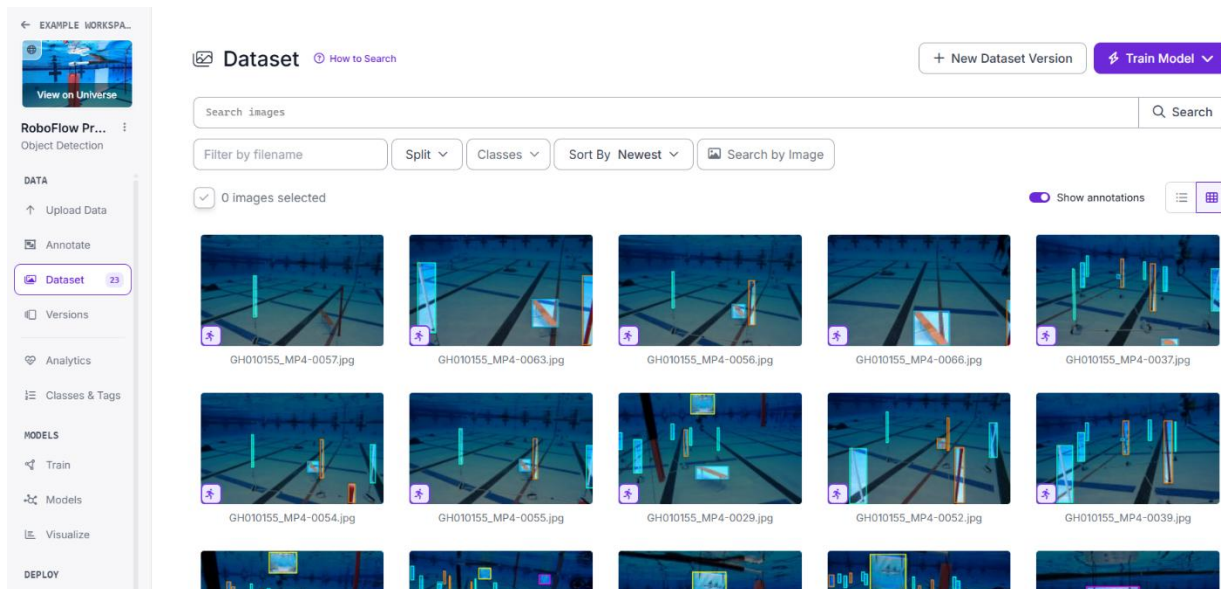


Figure 15 Dataset

38. Don’t select “Train Model” in right hand corner of Figure 15. We need to create a “New Version” before we can train a model, to keep things organized. Select “New Dataset Version”.
39. After selecting “New Dataset Version”, it will prompt you for some settings in your version. First of all, give it a descriptive name (maybe “1<sup>st</sup> Model Training”). Then, select edit under “Train/Test Split”. Select “Rebalance” and adjust the slider. The slider looks like Figure 16.

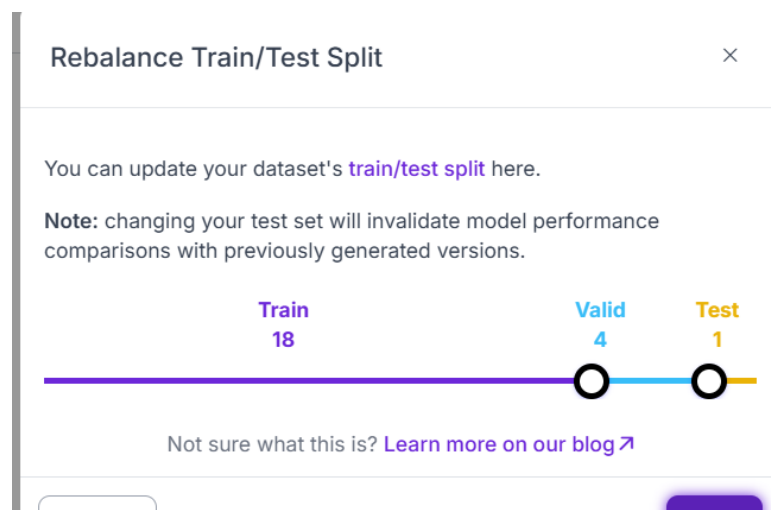


Figure 16 Image Balance

40. There are only 23 images in my example, but you will have 175. Slide the bar until you get Train = 140, Valid = 26, and Test = 9. This corresponds to a 80% = Train, 15% = Validate, and 5% for Test, which is pretty standard. Every other setting is OK. Create your version.

41. After creating a version, your screen should look like Figure 17.

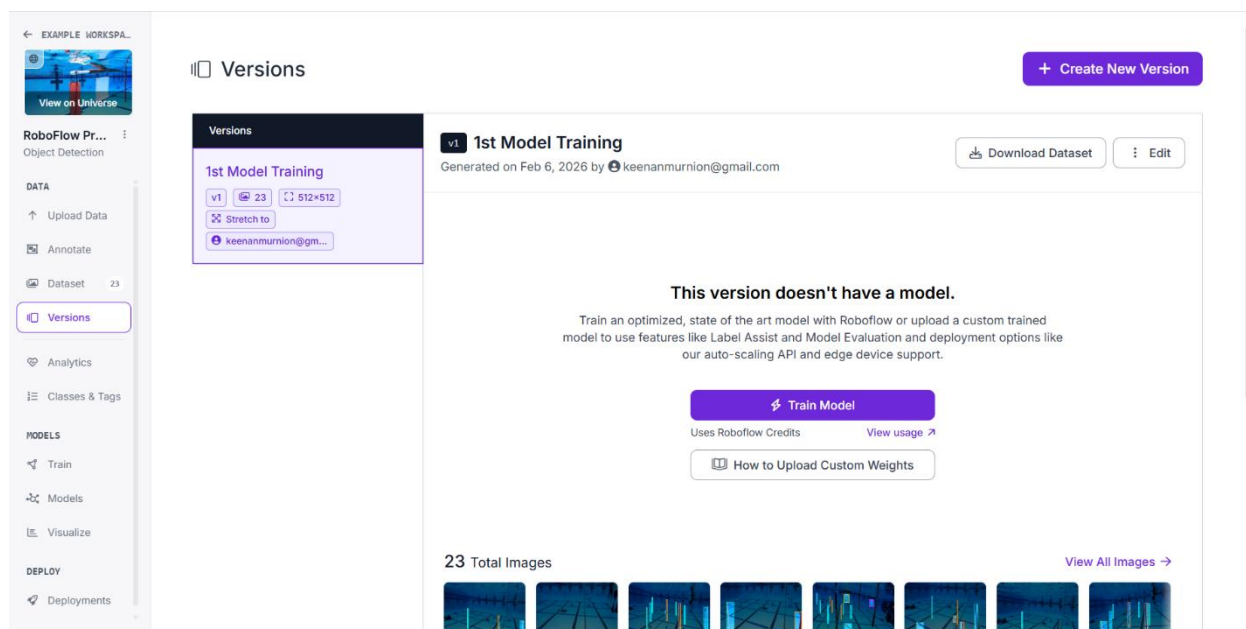


Figure 17 New Version Created

42. Now, it is time to train a model. Select the “Train Model” button that you can see in Figure 17.

43. You will get the option to choose 4 different architectures. Choose “YOLOv11”. This architecture has the best qualities (compared to the other options) of training time, inference time, and model accuracy.

44. After clicking on YOLOv11, you will be asked which version do you want to train on. If you have directly followed this example, simply choose Version 1. However, if you develop multiple versions over the course of using Roboflow multiple times, you will need to be careful at this step. Take care to select the correct version.

45. Select “Train Model”. Roboflow will give you an estimate as to how long model training will take, and then tell you that an email will be sent to you when it is done.



46. Your screen should look like Figure 18 when training.

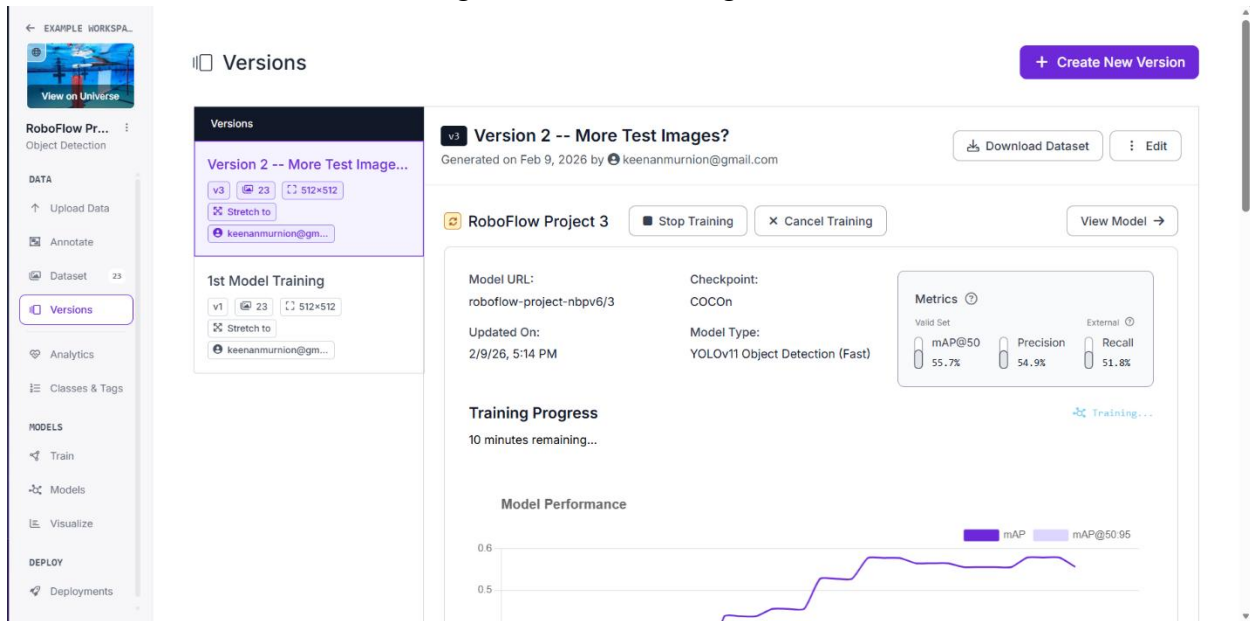


Figure 18 Model Training

47. Please note in Figure 18, I had to make a second version. This was because you need more than 1 image allocated to testing before you can create a model, which is what I had in “1<sup>st</sup> Model Training”. You will not have this problem, but I will document it.

48. The training that I did for my 25 images in the creation of this procedure took around 15 minutes. I have personally created a model with 175 (and more) training images, and can estimate that it will take between 30-50 minutes, depending on internet connection.

49. At this point, wait for the model to finish training.

50. At this step, the model should be trained. You can verify that it has trained by looking in the “Versions” tab, and seeing that model training has stopped. You can also look in the “Models” tab. You should see something like Figure 19.

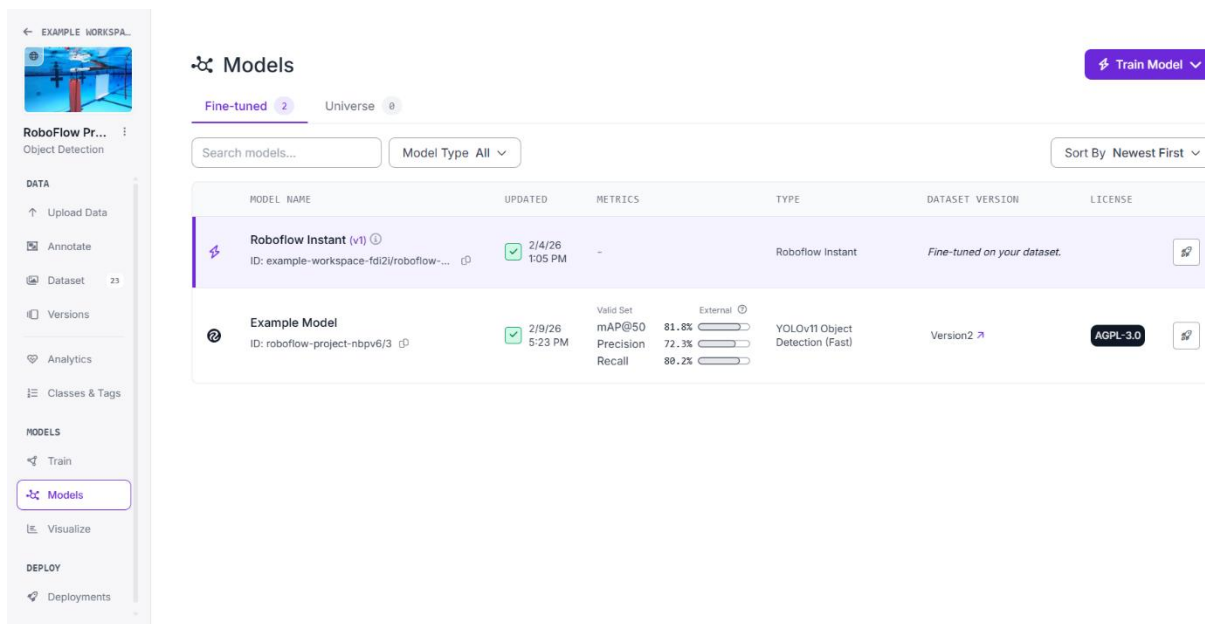


Figure 19 Models

51. Roboflow automatically trains a “Roboflow Instant” model. Disregard this. You can see from Figure 19 that I renamed the model I trained “Example Model”. Like anything, it is good to give your model a distinctive name so you don’t get later iterations confused.

52. It’s time to label the rest of the dataset!

53. Back in the “Annotate” tab, your screen should look like Figure 20.

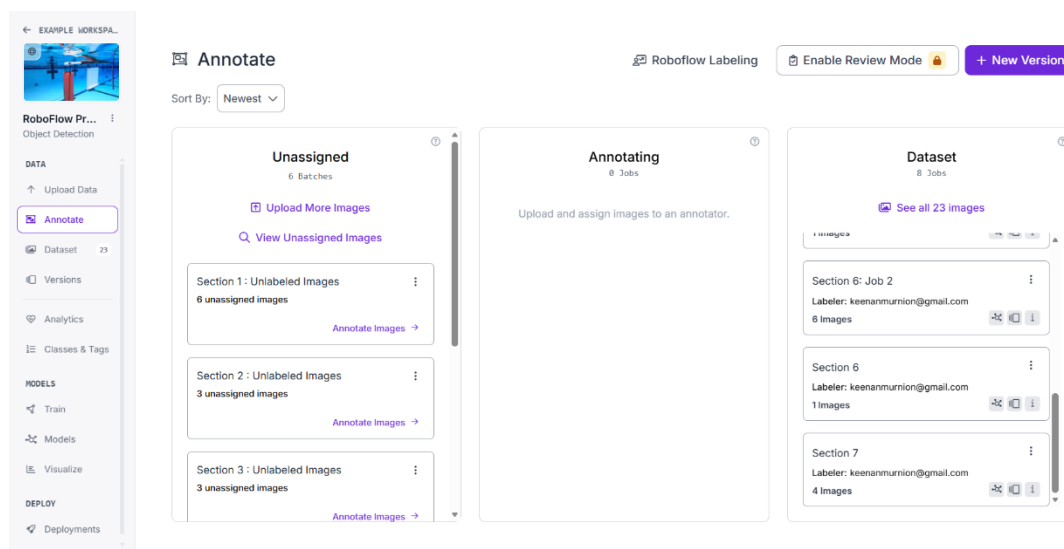


Figure 20 Annotate Tab

54. There should be 7 batches under the Unassigned tab, filled with the rest of the dataset images. We are going to label them all, and put them in the “Dataset” column. Just so we are clear, you should then have 14 batches in your Dataset column when you are finished labeling.
55. Click on “Annotate Images” under the “Unassigned” column for any of your remaining batches. This will be exactly like Step 21 and Figure 7. I am not going to reference them again here. However, instead of selecting “Label Myself”, you will need to select “Auto-Label Entire Batch”. This will allow you to deploy your model on the rest of the images.
56. Your screen should look like Figure 21.

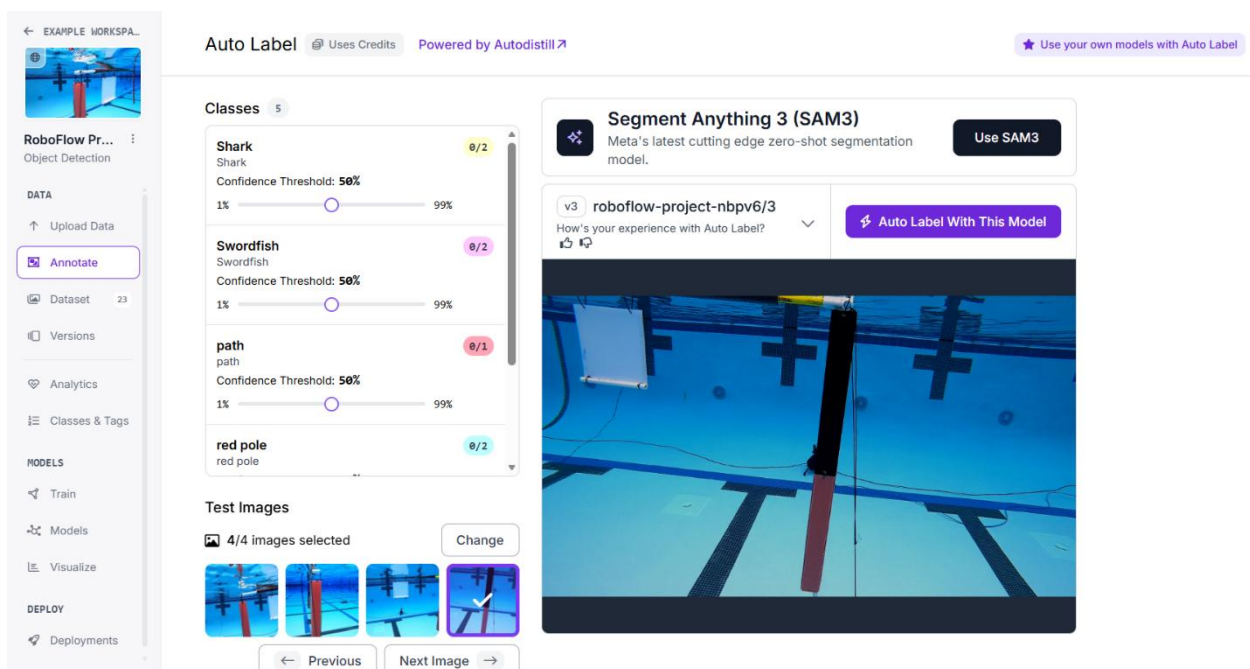


Figure 21 Labeling with Model

57. Referring to Figure 21, you can select which model you label with on top of the big image. You should have only 1 model if this is your first trial, but later iterations can select different models here. Also, set the Confidence Threshold slider bar on the left side of the Figure to 40% for every class. This has shown to produce the best results. It defaults to 50%, which is what Figure 21 has, but you must change it.
58. On the screen of Figure 21, RoboFlow will generate examples of the model labeling on 4 images in your dataset. The images I am using for my example dataset do not have any objects in them that I want to label. Thus, the model correctly did not label anything in them. You will see some model predictions when you do this step. Select “Auto Label With This Model”.

59. Auto labeling has taken me around 45 seconds in the past, even on large batches. If it is the first batch you are labeling with the model, it might take longer initially. Five minutes is the maximum time I would allow it to continue generating labels until I exit Roboflow and relaunch the program. It will be faster the second try. It seems to have issues, rarely, with the first batch.

60. After the model labels, your screen will look Figure 22.

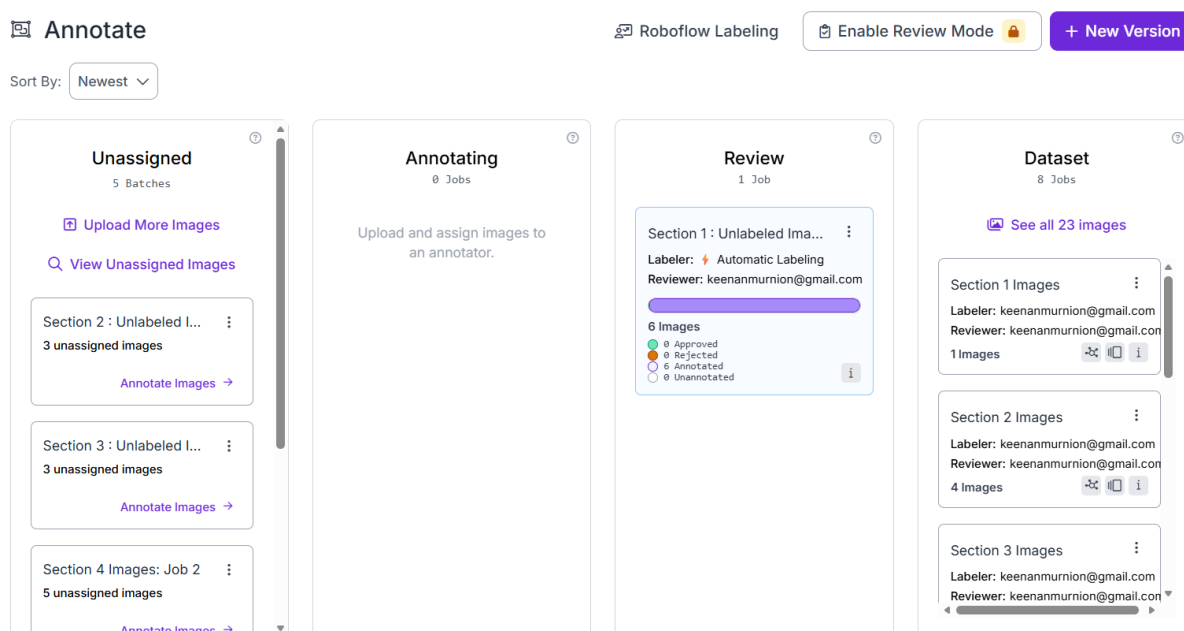


Figure 22 Review Model Labeling

61. Notice how there is now a “Review” column. This is because no labeled images will be put into the “Dataset” column without explicit approval from the user. **Don’t put any images into the Dataset that aren’t useful (labeled correctly).**

62. Go through the images the model labeled. Roboflow will give you the option to correct any potential errors, and then approve or reject any image.

63. Go through each image. Fix any mistakes you want, bounding box heights, incorrect labels (rare), draw more bounding boxes. After correcting mistakes, select the green “Approve” button near the top of the labeling screen. Reject any images that do not contain any labelable images.

64. The model does not perform perfectly. The point of creating the model was to rapidly label 100's, or even 1000's, of images quickly. Going through and fixing the slight mistakes of the model is much, much quicker than labeling images manually. That is the whole point of this step in creating the computer vision model.
65. Label the rest of the batches. Approve all images into the Dataset. This step will take the longest of any in this procedure. Checking the labels of the images is a tedious task, though less tedious than labeling them yourself! In prototypes, the model has performed exceedingly well, and reduced time by 20% on small datasets. I could extrapolate the results to expect a 50-70% time decrease on large competition datasets. **Depending on how many images your project has, this can take between 3-5 hours.**
66. Once everything is labeled, the Dataset should have all of the labeled images (manual, model labeled, model labeled with user corrections) in it. Your "Annotate" tab should have 14 batches under the "Dataset" column.
67. All labeled images are in the Dataset. We must now create a new version, so that you can download the images to both augment them, and train and validate a model on Tempest (covered in next procedure sections).
68. As you create a new version, **you must set the train/valid/test splits accordingly!**
69. Set the "Train" images to 80% of the total dataset. As there will be a different number of images, you will need to calculate what this number is. Do 20% for "Valid". To create a version, you must have at least 1 image for "Test".
70. For example, if you have 3000 images, set the slider to have 2400 images to "Train", 599 images to "Valid", and 1 image to "Test". Name your version accordingly. I suggest "Final Labeled Images". Create your version.
71. You have completed this section! In the next sections, you will download this version, and see that the "Train" and "Valid" images are in separate folders upon download. You can use each set of images accordingly.

**This concludes the Competition Workflow for Data Preparation**

## Model Creation – Competition Workflow

At the RoboSub competition, there is limited time to train the model. Before competition, users are required to fully run through the setup section to make sure their settings are correct, and that they are signed in properly to Tempest, GitHub, and Roboflow.

The following section lists the ordered steps on how to train a YOLOv8 model in Tempest. This section must proceed after the “Data Preparation” section. The user must have an image dataset already labeled in Roboflow in order to train a model off of the labeled objects.

1. This procedure goes through steps to train a model in Tempest. The first step is to access and log into Tempest from your device. If you have already done this, skip to step 8.
2. Sign into your Cisco VPN, if you have completed this go to step 4. On a Windows device this can be accessed by the ^ in the bottom right corner, next to your date and time symbols, shown in Figure 10:

### Connect:

1. Open **Cisco Secure Client**.

The app can be found by clicking the small white up arrow in desktop tray and then clicking Cisco Secure Client globe-icon.



*The Ready to Connect VPN dialog box opens.*

2. Enter **vpn.msu.montana.edu** (if not already entered) and click **Connect**.
3. Log in:
  - a. Choose **Group** appropriate to your role at MSU (in most cases this will be **MSU-Employee-VPN** or **MSU-Student-VPN**)
  - b. Enter your **NetID** and **NetID Password**.
  - c. Click **OK**.

Figure 21 Sign into Cisco VPN

3. Sign into the dialog box using your NetID email, Eg: a12b345@student.montana.edu. And your student password, Eg: Abcd\_1234.

4. Once connected to the VPN, go to the Tempest website in your browser at this link:  
<https://tempest-web.msu.montana.edu/>
5. Your screen will look like Figure 11.

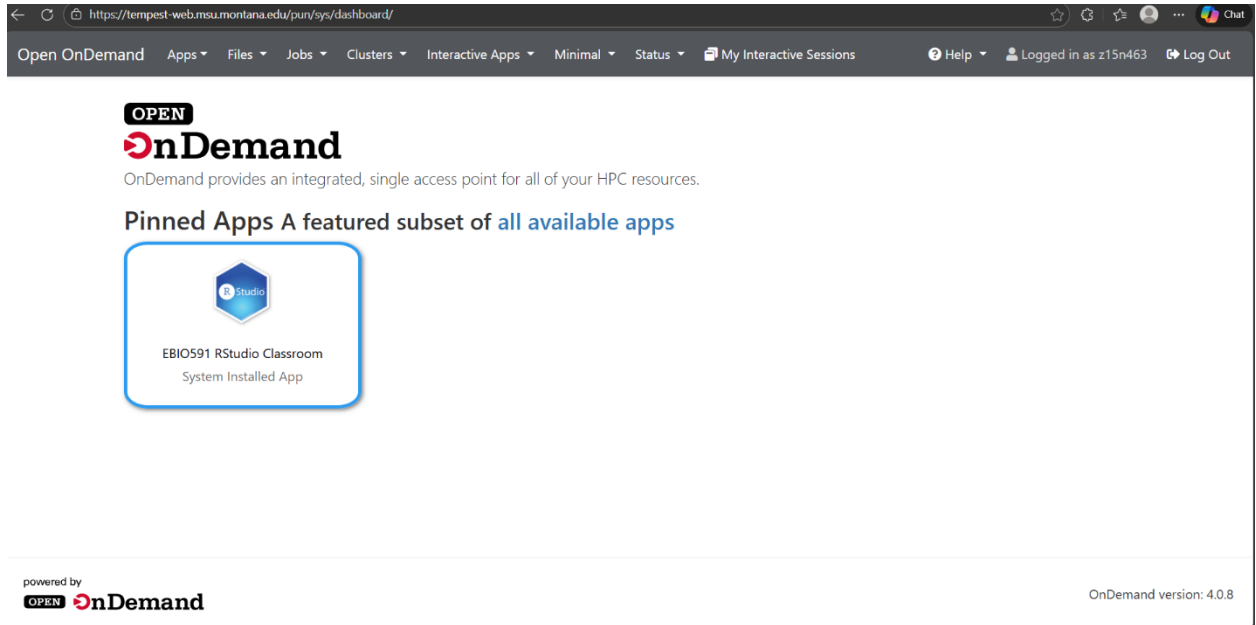


Figure 22 Tempest Home Screen

6. Select the “Files” dropdown menu from Figure 11, and select “Home Directory” from within the drop down options. This will take you to your home directory.
7. To get to the terminal, select the “>\_ Open in Terminal” button that appears in your home directory screen. Your screen should now look like Figure 12.

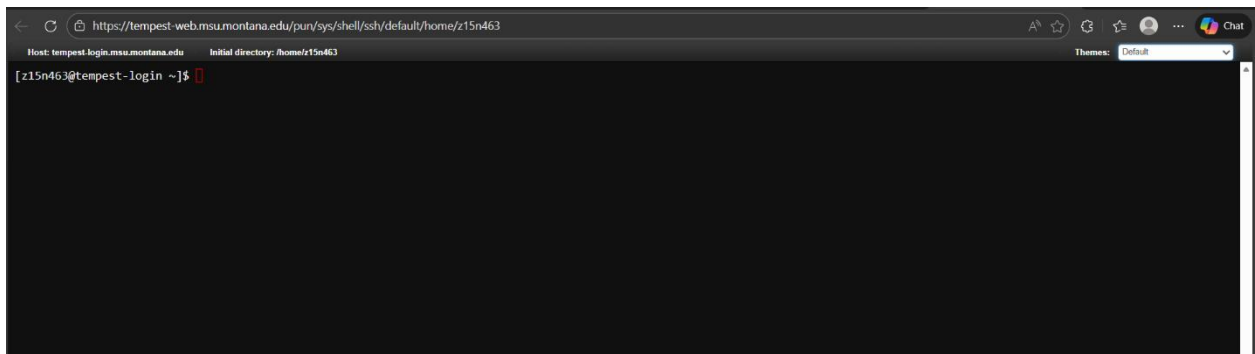


Figure 23 Tempest Terminal

8. Once you are in the Tempest terminal, navigate to the RobosubCV repository by typing:  
`cd RobosubCV`
9. In order for the program to know what dataset to download, you need to give the ‘train.sbatch’ script the URL. If you have done this, or are re-using the dataset, skip to step 15. Otherwise, log into [roboflow.com](https://roboflow.com) on your browser, and select the dataset, and the version you want to train. Figure 4 shows what your screen should look like to select the right dataset:

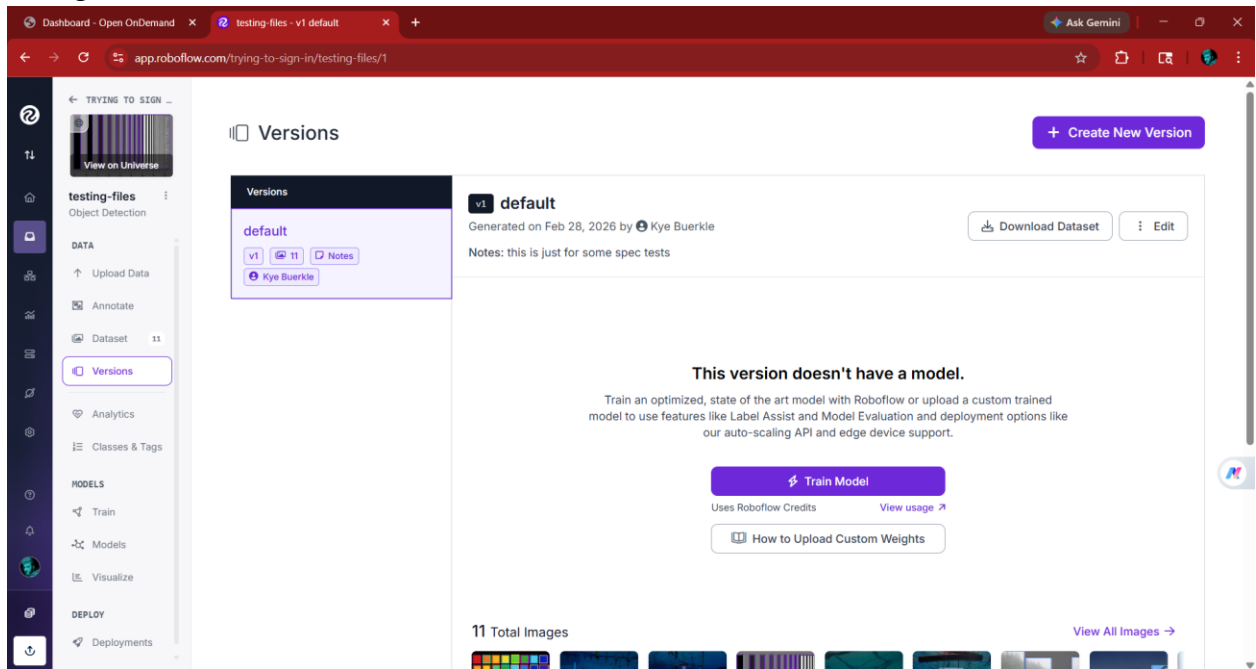


Figure 24 Roboflow dataset URL

10. Go to the address bar, and copy the URL of the dataset, Eg from Figure 24:  
<https://app.roboflow.com/trying-to-sign-in/testing-files/1>
11. Go back to your Tempest tab, and open the ‘train.sbatch’ script in a text editor:  
`nano sbatch/train.sbatch`



12. The script should look like this, (second picture is a zoom in on the configuration):

```
Host: tempest-login.msu.montana.edu Initial directory: /home/v67n884
GNU nano 2.9.8 train.sbatch

#!/bin/bash

priority-bradwhitaker or group-bradwhitaker
priority faster for gpupriority (2 A100 or 2 A40); H100 nextgen-gpu (4 H100 and 3 nodes) faster, but wait longer
#SBATCH --account=priority-bradwhitaker
#SBATCH --partition=gpupriority
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=4
#SBATCH --gpus-per-task=a40:2
1000G max for H100
500GB for Quad A100
#SBATCH --mem=128G
#SBATCH --time=0-06:00:00
#SBATCH --job-name=RobosubCV-full-1
%x is the job name and %j and the job number
#SBATCH --output=./stream-out/%x-%j.out
#SBATCH --error=./stream-out/%x-%j.err

module load Anaconda3/4.0.0
source activate $HOME/.conda/envs/test-train

Past the URL to your dataset here
ROBOFLOW_URL=[ROBOFLOW_URL]

poetry install
poetry run python $HOME/RobosubCV/src/model_training/train.py -u $ROBOFLOW_URL

Past the URL to your dataset here
ROBOFLOW_URL=[ROBOFLOW_URL]
```

Figure 25 train.sbatch script

13. Delete the “[ROBOFLOW\_URL]”, or any url that has been pasted in that spot next to “ROBOFLOW\_URL=”.

Paste your url in there. (Note: there can’t be any spaces)

(Note: nano doesn’t allow use of a curser, only arrow keys)

It should look like Figure 26:

```
Past the URL to your dataset here
ROBOFLOW_URL=https://app.roboflow.com/trying-to-sign-in/testing-files/1
```

Figure 26 Roboflow URL in script

14. Save the file, nano uses “ctrl + s” and then “ctrl + x” to exit.

15. Now you can run the script in Tempest to train the model on that dataset. Run this command:

```
sbatch sbatch/train.sbatch
```

16. On success you should see this:

```
[v67n884@tempest-login ~]$ cd RobosubCV/
[v67n884@tempest-login RobosubCV]$ nano sbatch/train.sbatch
[v67n884@tempest-login RobosubCV]$ sbatch sbatch/train.sbatch
Submitted batch job 3716160
[v67n884@tempest-login RobosubCV]$
```

That submits the request to use Tempest with the given job id.

You can now close out of the Tempest terminal if you wish.

17. In order to track the progress of the model, go to the Tempest website as listed in step 4 and go to Jobs → Active Jobs, to see the current progress of this job.

- a. In queue, means that Tempest is waiting for an empty spot to run the program, so currently nothing is happening.
- b. In progress, means that Tempest is currently running, and training your model.
- c. Complete, means that the job has been completed and you can move onto the next step
  - i. If the job completes in under a minute, it defiantly failed, troubleshoot the issue before trying again (troubleshooting section below)
- d. If your job was running, or queued, and then later it disappeared from your jobs list, it means it completed. To check out

18. Once completed, check that the model completely finished by navigating to your Results Directory. (This is determined by the configuration.json from step 42 in the setup section above)

- a. In Tempest click on RobosubCV → results

19. Make sure there is a new “train” folder, and that there is a “weights” folder in there, and a “best.pt” or “last.pt” file in there. train → weights → best.pt

- a. This .pt file is the model file you will need to upload in the next section of the procedure: Model Upload
- b. Each new “train” folder adds a number to the end, the newest folder has the highest number, that is most likely your new model folder
- c. If this didn’t work, or you don’t know if this is a new train folder, then go to the Troubleshooting section for Model Creation (below)

20. Select the “best.pt” file, and press the download button in the top right corner of the screen. This will download the model into the “Downloads” on your device.
21. You are finished with model creation! To upload this to the RoboCats UAV, go to the Model Creation section.
  - a. Before you upload to the submarine, you can verify the accuracy of the model in the **Parameters and Accuracy** section at the end of this procedure.

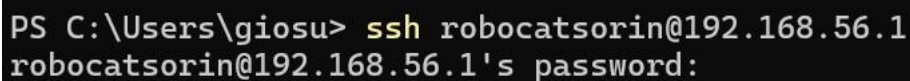
**This concludes the Model Creation Procedure Section.**

## Model Upload – Competition Workflow

This section covers how to transfer the trained computer vision model from your laptop to the submarine's onboard computer (Jetson Nano). You must have completed the Model Upload Setup section and have the Ethernet cable connected between your laptop and the submarine before proceeding. This process takes approximately 5-10 minutes.

1. SSH connection to Robosub
  - a. Open 'Windows PowerShell'
  - b. Form a SSH connection to the Robosub using the following command:

```
ssh robocatsorin@192.168.56.1
```



```
PS C:\Users\giosu> ssh robocatsorin@192.168.56.1
robocatsorin@192.168.56.1's password:
```

- c. When prompted for the password enter (it will not show that you are typing, but you are): R0boCats!
    - d. Once successfully logged into the Robosub navigate to the correct directory for the model to be using the following command

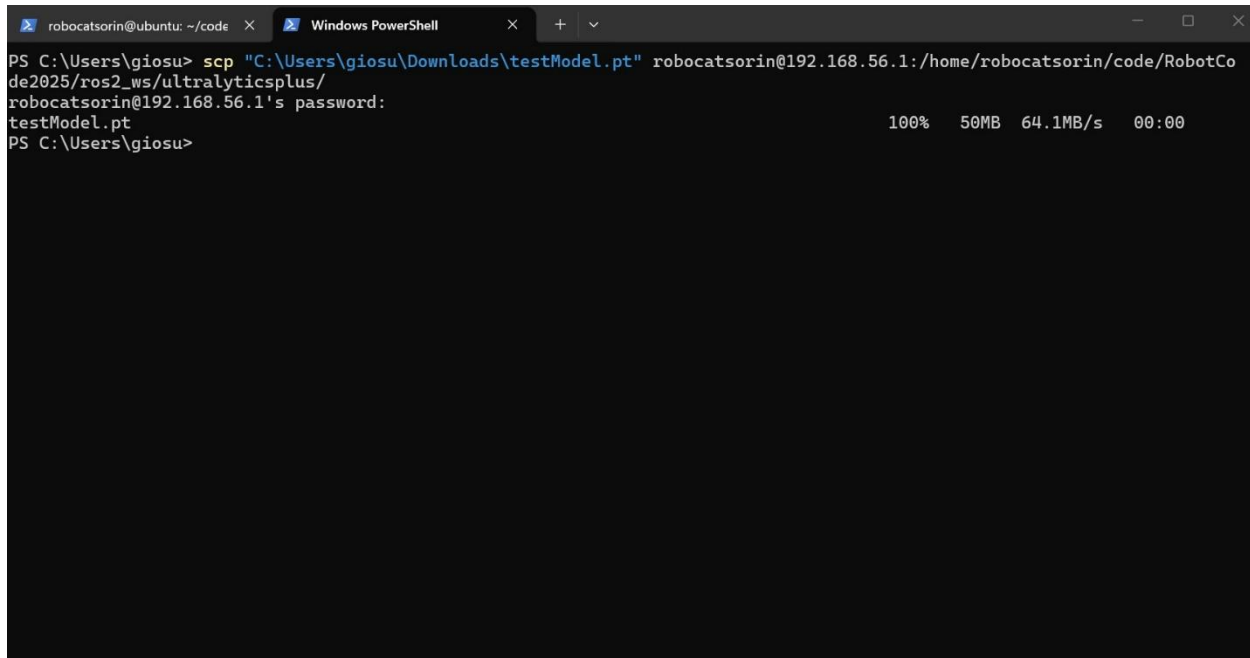
```
cd code/RobotCode2025/ros2_ws/ultralyticsplus/
```



```
robocatsorin@ubuntu:~$ cd code/RobotCode2025/ros2_ws/ultralyticsplus/
robocatsorin@ubuntu:~/code/RobotCode2025/ros2_ws/ultralyticsplus$ |
```

- e. Open a new terminal in Windows PowerShell, but do not close the first terminal.
      - f. In the new Windows PowerShell terminal, upload the YOLOv8 model using the following command. Replace "<username>" with your laptop's Windows username and "<modelname>" with the name of your model file. This must be entered as one single lines, and remember to include the quotation marks.

```
scp "C:\Users\<username>\Downloads\<modelname>.pt"
robocatsorin@192.168.56.1:/home/robocatsorin/code/RobotCode2025/ros2_w
s/ultralyticsplus/
```



```
robocatsorin@ubuntu: ~/code X Windows PowerShell X + v
PS C:\Users\giosu> scp "C:\Users\giosu\Downloads\testModel.pt" robocatsorin@192.168.56.1:/home/robocatsorin/code/RobotCo
de2025/ros2_ws/ultralyticsplus/
robocatsorin@192.168.56.1's password:
testModel.pt 100% 50MB 64.1MB/s 00:00
PS C:\Users\giosu>
```

- g. When prompted, re-enter the password: R0boCats!
- h. Wait for the file transfer to complete. The terminal will show the upload progress.
- i. Go back to the first PowerShell window (the one connected to the submarine). Type the following command to verify the model was uploaded successfully: `ls`
- j. You should see your model file name in the list. If it appears, the upload is complete.

After confirming the model file is present on the submarine's computer, the model upload is complete! The submarine is now ready to use the new computer vision model for competition. Close both PowerShell windows and disconnect the Ethernet cable. The computer vision software development process is now finished!

**This concludes the Model Upload Procedure, and the Competition Workflow section of the procedure.**

# Summary and Troubleshooting

This concludes the RoboSub Computer Vision Software Development Procedure. By following the steps in this document, the RoboCats team has collected competition object footage, labeled the images, augmented the training data, trained a computer vision model, and uploaded it to the submarine. If any issues were encountered during the process, refer to the troubleshooting guides below.

## 1. Data Collection Troubleshooting

Common Mistakes:

- Entering the wrong dataset root directory when prompted by the script
- Using spaces in folder or file names
- Forgetting to create object class folders inside renamed\_sorted before sorting videos
- Running the script before videos have been copied into the correct class folders
- Assuming all objects is rebuilt automatically after extraction, pruning, restore, or delete actions
- Skipping option 9 after cleanup and then labeling or exporting an outdated all\_objects folder

Expected Result:

- Videos are sorted by object class in renamed\_sorted
- Renamed videos are created by the script
- Extracted 640x480 frames are saved in 640x480\_frames
- all\_objects is rebuilt with the current kept frames after running option 9
- Each object class is capped to 500 frames before labeling

## 2. Model Creation Troubleshooting

This shows a couple possible problems, and their potential solutions when creating a yolov8 model using the RobosubCV repository.

If you want to decrease the time of training, or check the accuracy of the model and run it on a camera, go to the **Parameters and Accuracy** section below.

For more details in on how to use the repository, read through the README.md documents in each directory: <https://github.com/kyebuerkle/RobosubCV>

For more detail on how to use Tempest, go to the MSU Tempest website:

<https://www.montana.edu/uit/rci/tempest/>

For more details on how to use Linux commands, here is a helpful website:

<https://www.geeksforgeeks.org/linux-unix/basic-linux-commands/>, or look up “basic linux commands” on your browser.

For more details on how to use git commands, look up “git commands” and go to GIT CHEAT SHEET from [education.github.com](https://education.github.com), or any other website.

If you don’t see a “train” folder in your results, or you don’t have a .pt file to upload, the best way to troubleshoot any issues is by reading through the “stream-out” files Tempest outputs while training. To access these, go to your Tempest homepage, and navigate to: RobosubCV → stream-out (if that doesn’t exist go to: RobosubCV → sbatch → stream-out). Find the 2 files labeled “RobosubCV-1234567.err” and “RobosubCV-1234567.out” where 1234567 is your job id that was output in the terminal when you submitted the job (they are numerical, so the largest number of the bunch is the most recent). Look through the .err file first, and then the .out file. .err stands for the error stream out, but to debug any issues, it is usually printed to the .out file.

Common .err messages:

This is a list of .err messages that are completely fine and expected.

- `/home/v67n884/.conda/envs/Training/lib/python3.12/site-packages/torch/utils/data/dataloader.py:424: UserWarning: This DataLoader will create 8 worker processes in total. Our suggested max number of worker in current system is 4, which is smaller than what this DataLoader is going to create. Please be aware that excessive worker creation might get DataLoader running slow or even`

freeze, lower the worker number to avoid potential slowness/freeze if necessary.  
self.check\_worker\_number\_rationality()

This warning is fine for YOLO, this doesn't effect the training process.

```
- Extracting Dataset Version Zip to /home/v67n884/RobosubCV/data/2025-robosub-on-site-17qeg-moyor-v2 in yolov8:: 94%| | 5896/6248 [00:15<00:00, 524.83it/s]
Extracting Dataset Version Zip to /home/v67n884/RobosubCV/data/2025-robosub-on-site-17qeg-moyor-v2 in yolov8:: 95%| | 5952/6248 [00:15<00:00, 426.06it/s]
Extracting Dataset Version Zip to /home/v67n884/RobosubCV/data/2025-robosub-on-site-17qeg-moyor-v2 in yolov8:: 96%| | 6023/6248 [00:15<00:00, 494.50it/s]
Extracting Dataset Version Zip to /home/v67n884/RobosubCV/data/2025-robosub-on-site-17qeg-moyor-v2 in yolov8:: 97%| | 6084/6248 [00:15<00:00, 524.06it/s]
Extracting Dataset Version Zip to /home/v67n884/RobosubCV/data/2025-robosub-on-site-17qeg-moyor-v2 in yolov8:: 98%| | 6153/6248 [00:15<00:00, 568.41it/s]
Extracting Dataset Version Zip to /home/v67n884/RobosubCV/data/2025-robosub-on-site-17qeg-moyor-v2 in yolov8:: 99%| | 6214/6248 [00:16<00:00, 576.85it/s]
- Extracting Dataset Version Zip to /home/v67n884/RobosubCV/data/2025-robosub-on-site-17qeg-moyor-v2 in yolov8:: 100%| | 6248/6248 [00:16<00:00, 388.02it/s]
```

This shows a successful dataset download from roboflow

### Common .out messages:

When reading the .out, start at the bottom, since this will indicate if there was any errors to begin with. You only need to read through the entire file if the training failed, and you need to find out where it didn't succeed.

- The top of the file should look like this on success:

```
Installing dependencies from lock file

No dependencies to install or update

Installing the current project: robosubcv (0.1.0)
Dataset already exists, Do you wish to overwrite?

You are already logged into Roboflow. To make a different login,run roboflow.login(force=True).

loading Roboflow workspace...

loading Roboflow project...

Ultralytics 8.4.14 🚀 Python-3.12.12 torch-2.10.0+cu128 CUDA:0 (NVIDIA A40, 45490MiB)
CUDA:1 (NVIDIA A40, 45490MiB)
```

- The bottom of the file should look like this:

```
Results saved to [1m/home/v67n884/RobosubCV/results/train9[0m
Training success!
```

This shows you the directory that the “weights” are saved to and that it was a success (v67n884 is an example NetID, yours will be shown instead)

NOTE: the directory is between the “?[1m“ and the “?[0m”, to find it: Tempest starts at the “home/v67n884” section, so you will need to navigate to Home → RobosubCV → results → train9 → weights, or whatever “train#” your output ended up being.

### Common Issues

Here is a list of potential issues you may encounter:



```
[v67n884@tempest-login RobosubCV]$ poetry run python src/model_training/setup_settings.py
The currently activated Python version 3.11.7 is not supported by the project (^3.12).
Trying to find and use a compatible version.
Using python3.12 (3.12.11)
Skipping virtualenv creation, as specified in config file.
Traceback (most recent call last):
 File "/home/v67n884/RobosubCV/src/model_training/setup_settings.py", line 6, in <module>
 import roboflow
ModuleNotFoundError: No module named 'roboflow'
[v67n884@tempest-login RobosubCV]$
```

This image shows an example of the “ModuleNotFoundError” issue.

When running any script, you must be in your “Training” environment AND you must have the Poetry dependencies installed (It doesn’t hurt to check this twice either):

```
source activate Training
poetry install
```

On success it will look similar to this image, where “(Training)” is in the far left corner to verify your current environment, and the dependencies were successfully installed (if poetry doesn’t work, run: poetry install again, and if that doesn’t work, do the “If All Else Fails” troubleshoot below):

```
[v67n884@tempest-login RobosubCV]$ source activate Training
(Training) [v67n884@tempest-login RobosubCV]$ poetry install
Installing dependencies from lock file

No dependencies to install or update

Installing the current project: robosubcv (1.0.0)
(Training) [v67n884@tempest-login RobosubCV]$
```

- To leave your environment run the command: `conda deactivate`

This will bring you to the “(base)” environment, which is the same as home, if you really want the “(base)” to be gone use: `source deactivate` (but this is deprecated, and could be completely removed in the future)

- If the stream-out .out file had an error “yaml file not found at: ....” The ... is the directory to the dataset that the model is trying to train from. Navigate to that in tempest (each / is a folder). If the dataset doesn’t exist, or if inside is a .zip folder, it means Roboflow failed to download the dataset. (IDK why roboflow does this).

All you need to do is re-run the tran.sbatch by: `sbatch sbatch/train.sbatch`

Sometimes, the env\_setup.sh might not work (especially after some time, it might get out dated) these steps are the Legacy, and could still work to setup an environment

1. To create the `Training` environment, you need to run some commands from the terminal to setup the correct software environment on Tempest. Run these commands exactly:

```
cd $HOME
module load Anaconda3/2024.02-1
```

```
conda create -n Training python=3.12 -y
source activate Training
```

These commands set up a specific workspace (called an environment) within Tempest, with the correct software tools to train a computer vision model.

1. You should now see “(Training)” to the left of your username in the terminal, like Figure 6.

A terminal window with a black background. The prompt is "(Training) [z15n463@tempest-login ~]\$" in white text. A red cursor is visible at the end of the prompt.

*Figure 27 In the Training Environment*

2. Run these commands next, one at a time:

```
pip3 install --upgrade torch torchvision torchaudio
pip install poetry
poetry config virtualenvs.create false
```

These commands install torch dependencies and Poetry. “torch” is used to access and manage the GPUs in Tempest, using CUDA, for the python YOLO model to train on. Poetry is used in the RobosubCV repository to manage all other dependencies.

## If All Else Fails

If you can’t troubleshoot it, and figure out how to solve the issue you are running into, the best thing to do is to try again. You will keep all of your accounts, but start over the setup procedure.

1. The first thing is to reset the Training Environment. Run the command:

```
conda env remove --name Training
```

and start the setup procedure from step 18 to re-make the environment.

- a. You will still have your repository, so you can skip the steps that clone the repo
2. If your Repository is the issue, but you were able to clone it, run these commands:  

```
git checkout main
git pull
```

if this has an error similar to “you have things in your working directory” there are two things you can do

- to remove this (it will get rid of any changes you made) run:

```
git add --all
```

```
git stash
```

```
git pull
```

- to save any changes for later run:

```
git checkout -b [Branch Name you want to save it to]
```

```
git add --all
```

```
git commit -m "message you want to save"
```

```
git checkout main
```

```
git pull
```

Once you are in the main branch, run the setup from step 40